SPACE FACULTY

Let's learn about **satellites**!

Space mission and **satellite** design course

Space Mission Design Software

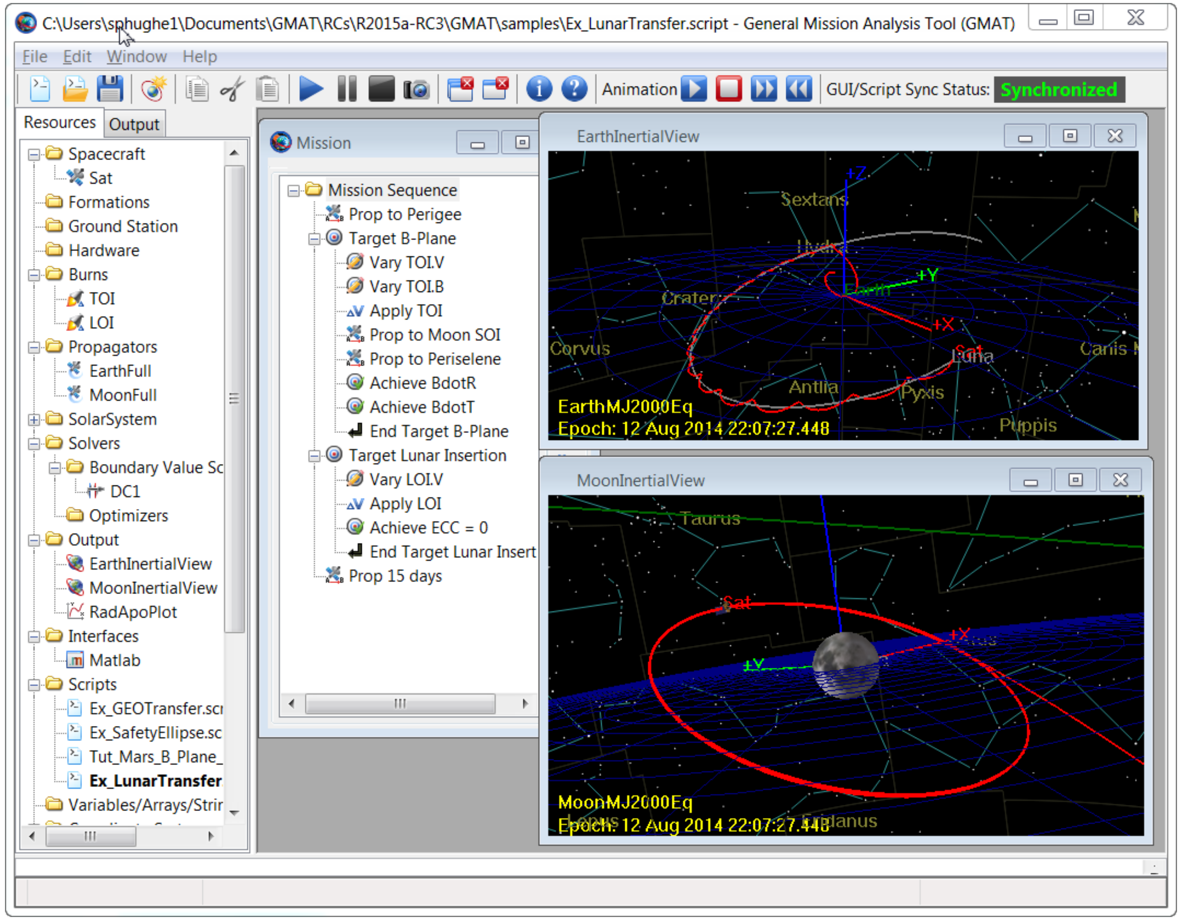
2NDSpace

What are Space Mission Design Software?

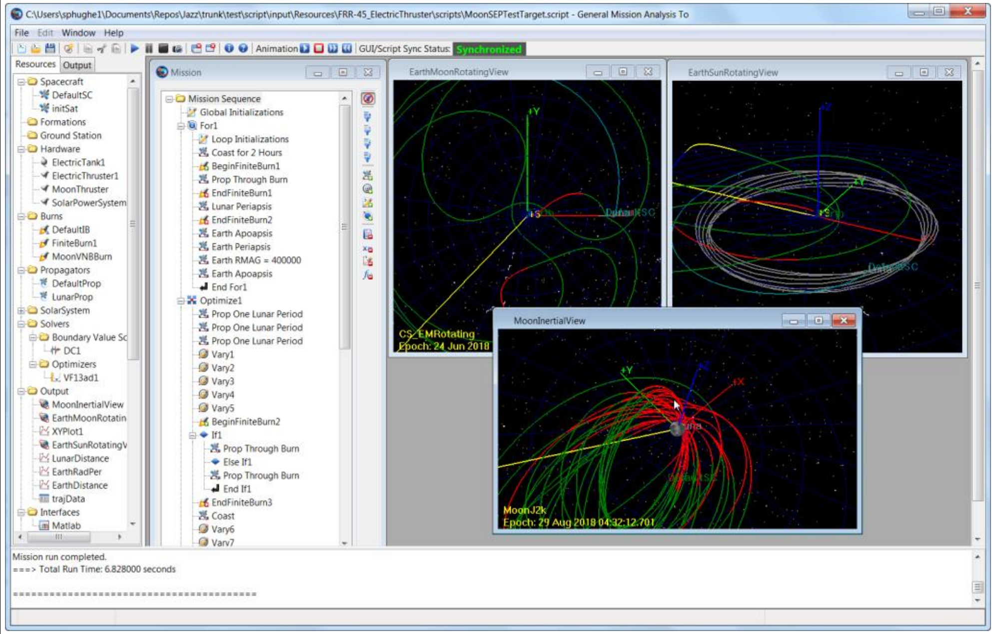
Designing space missions is hard.

For this reason, it is important to have some tools at our disposal to help during the design phase.

Mission Design Software are physics-based simulations that allow us to perform a variety of analyses from ground: we can compute how much fuel we need for a certain maneuver, check the ground plot of a satellite, even perform interplanetary missions!



When do we need these software?



We can check **before launch** what kind of orbit do we need to meet mission requirements, perform analysis on the **contacts** with ground stations, compute how much time of the orbit is spent in **eclipse** (and so without power generation from the solar panels).

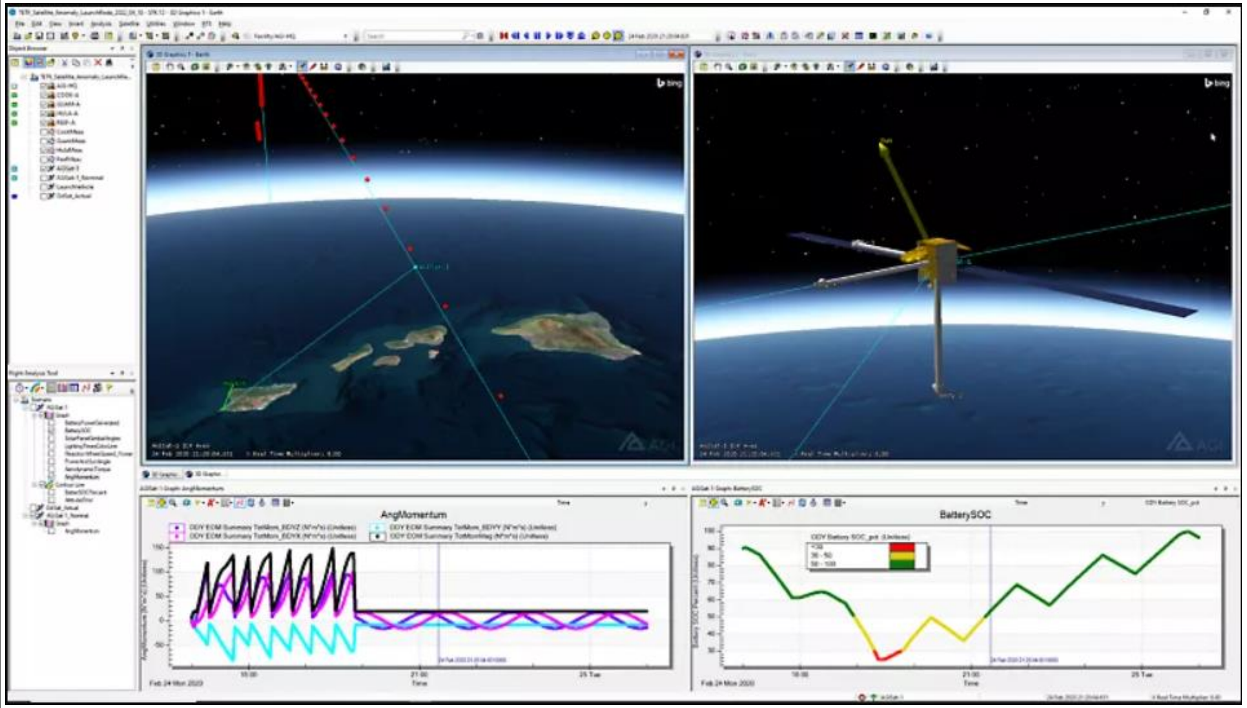
We can compute **after launch** when to perform a maneuver, and how much propellent we need.

Popular Software – GMAT

The General Mission Analysis Tool (GMAT) is an open-source space mission analysis tool. Open-source also means that it is **free**!

GMAT is designed to model, optimize, and estimate spacecraft trajectories in flight regimes ranging from low Earth orbit to lunar applications, interplanetary trajectories, and other deep space missions.

Popular Software – Ansys STK



STK is a very capable mission design software (and much more, it is used in a variety of different fields) designed by the company Ansys.

It is widely used in the industry.

Popular Software - FreeFlyer

FreeFlyer is a commercial off-the-shelf (COTS) software application for space mission design, analysis, and operations. FreeFlyer is used for spacecraft analysis and operations by various institutions and commercial satellite providers.

Popular Software

Orbitron is a satellite tracking system for radio amateur and observing purposes. It's also used by weather professionals, satellite communication users, astronomers etc. Application shows the positions of satellites at any given moment (in real or simulated time).

DAS The Debris Assessment Software (DAS) is provided by the NASA Orbital Debris Program Office as a means of assessing, during the planning and design phase, space missions compliance with NASAs requirements for reduction of orbital debris.

Popular Software – MATLAB, Python and so on!

Sometimes, a company decides to put up the work to use internally written software in order to assess specific problems.

In this case, a company could have full control over the software used and be conscious about every step of the calculation.

A hybrid approach is also often utilized: the data obtained from mission design software is later put into external scripts to perform specific analyses. We will see how to do it later!

We will learn about GMAT!

During this lesson, we will see how to use GMAT.

Being open-source, GMAT is **free to download and use.**

You can download it here.

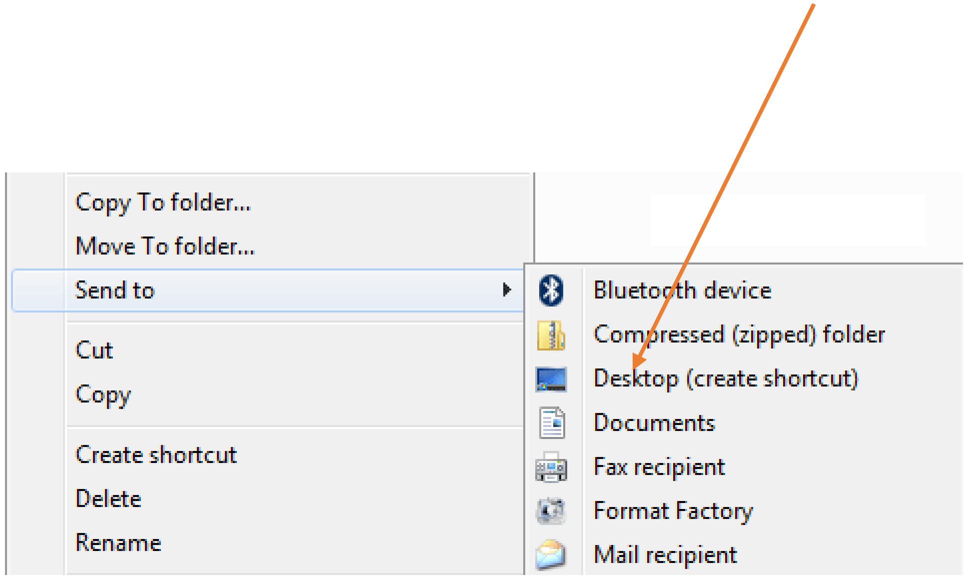
This means that you can have fun experimenting with the software yourself, after this lesson is over.

Installing GMAT - Windows

The GMAT Windows distribution contains a zip file. When unzipped,

locate the bin directory then select GMAT.exe to open GMAT.

You can also create a shortcut on desktop by right-clicking the program name and then click Send To > Desktop (Create Shortcut).



Installing GMAT - MacOs

The GMAT macOS distribution is now provided as a NASA-signed and notarized
DMG file, and is compatible with macOS 10.15 (Catalina) and above. This distribution was built and tested on macOS 11.7 (Big Sur).

After downloading and mounting (i.e. opening) the dmg file, installation of GMAT is as simple as copying the resulting "GMAT <version>" folder to one of the following two locations:

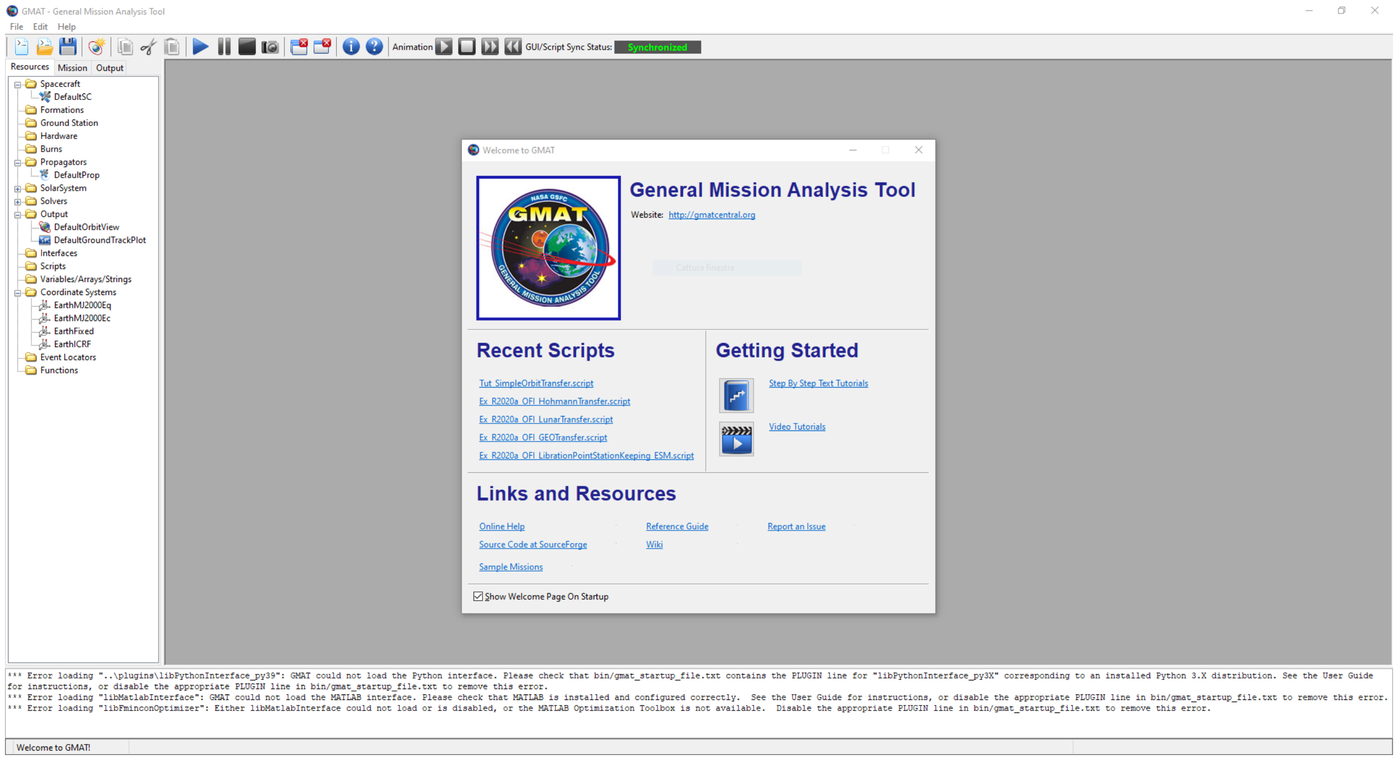
- System-wide /Applications folder. This requires admin privileges.
- User's $HOME/Applications folder. Admin privileges are not needed.

Copying GMAT to any other location will result in an error on launch.

GMAT – Welcome Page

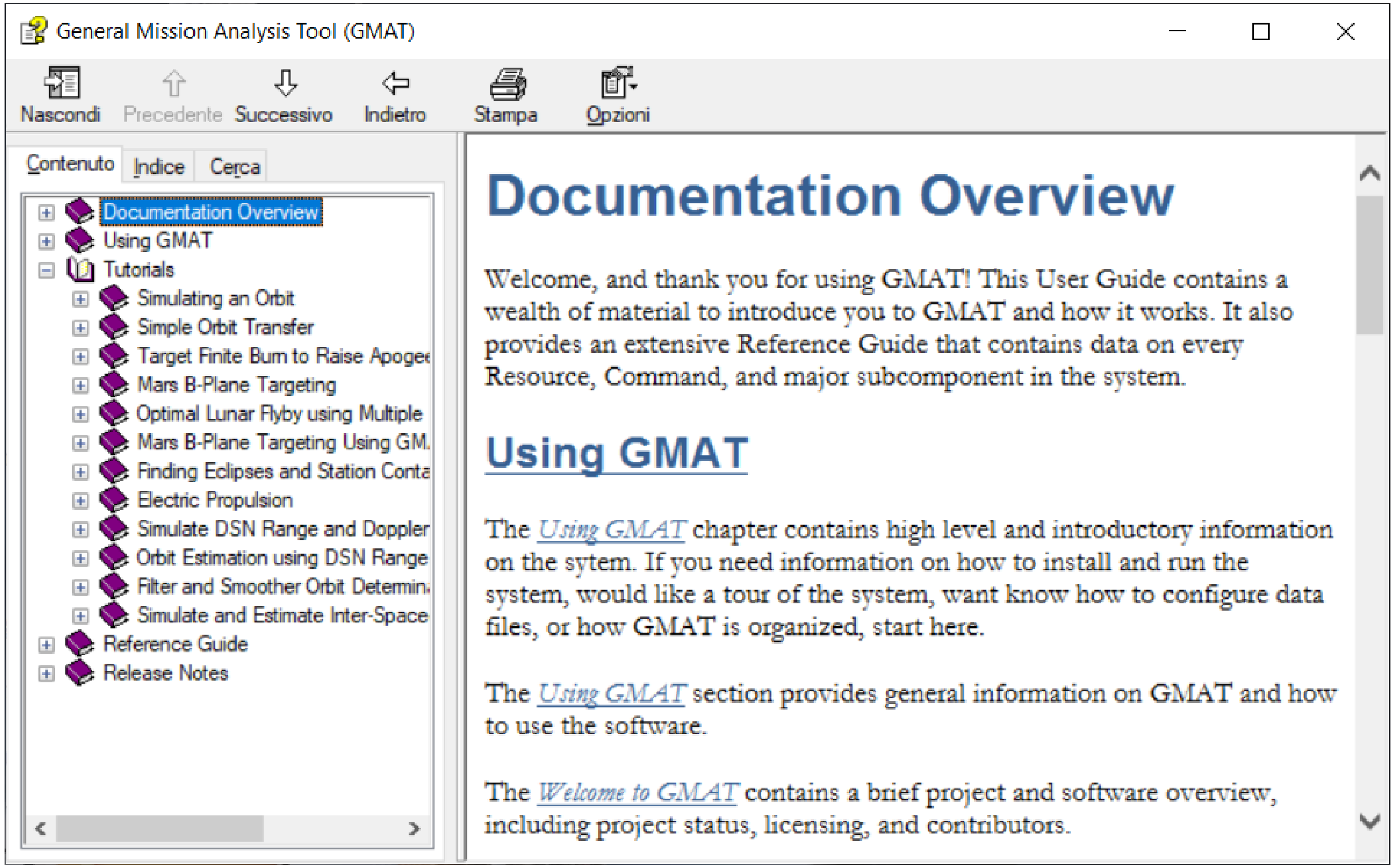
At your first opening, you can see a welcome page that show you the most recent scripts you worked with, a getting started section and some useful links and resources.

After this lesson, you can try to have a look at the provided tutorials if you want to learn more about GMAT.



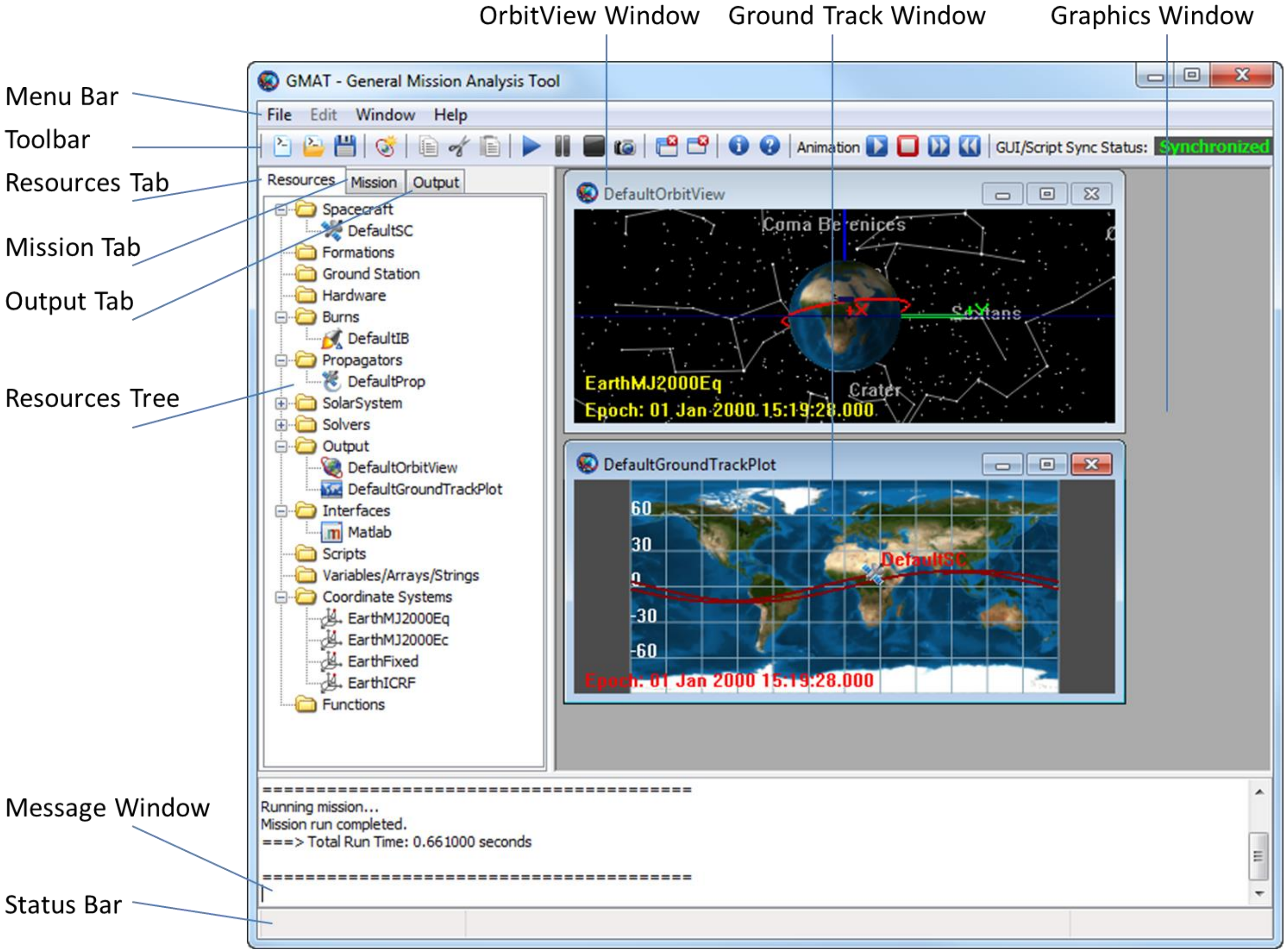
GMAT Documentation

If you click on the “Step by Step Tutorial” link on the welcome page, you will find a lot of useful resources to start learning. Today, we will do together a different tutorial from the one listed to highlight some of the things we have studied on previous lessons. But you could totally try the other tutorials!



GMAT Interface -1

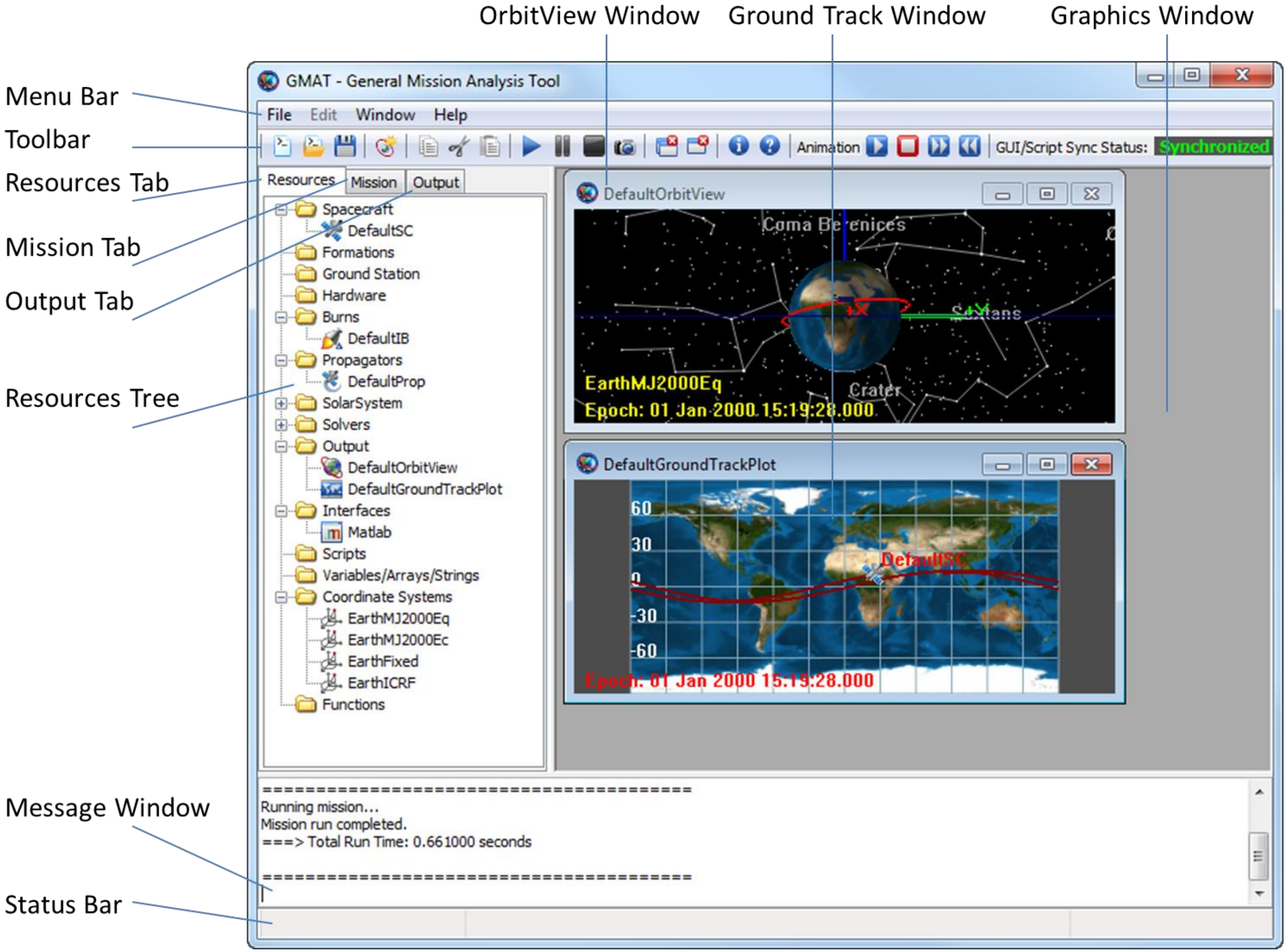
GMAT is driven by a GUI, or a Graphical User Interface. This means that you can operate the software by navigating around with your mouse and you have visual feedbacks and indications. In reality, the whole of GMAT is running in a script, a text document that experienced user can modify to work faster or for some specific applications.



GMAT Interface -2

First of all, GMAT is organized by three tab: the **resource**, **Mission** and **Output** tabs.

Over them, in the toolbar, you can save your project, run the simulation or control the animation. On the right there is a graphic window, where you can see the graphical output of your work.



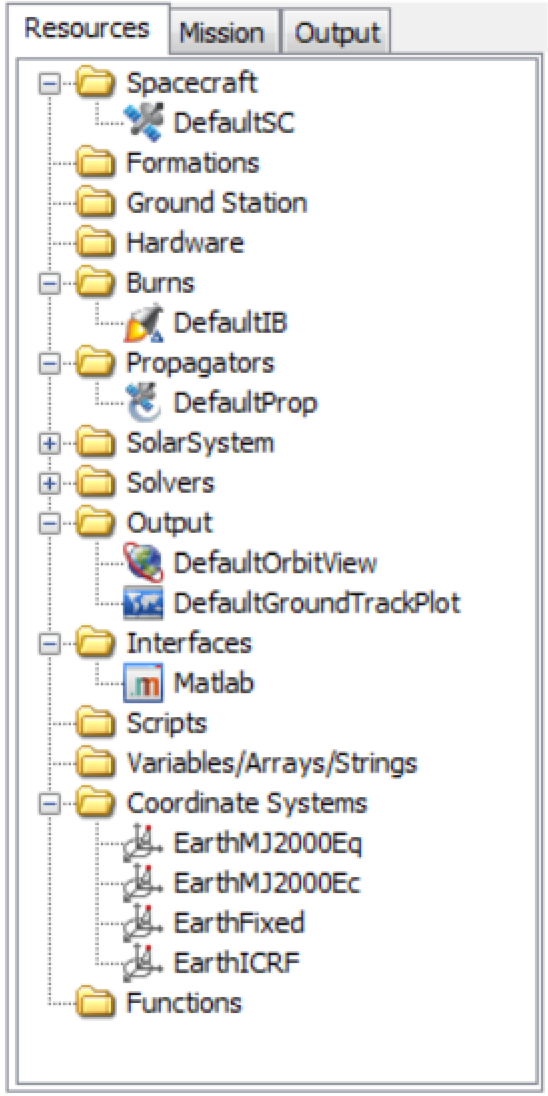
Resource Tab

In the Resource Tab you have at your disposal all the objects that are used in the simulation.

Using the Resource Tree, you can add, edit, or delete most of the resources.

For example, you can add a Spacecraft by right-clicking the Spacecraft folder, or you can edit anything by double-clicking on it.

Note that some Default object can not be deleted, like the default coordinate systems and the default planetary bodies.

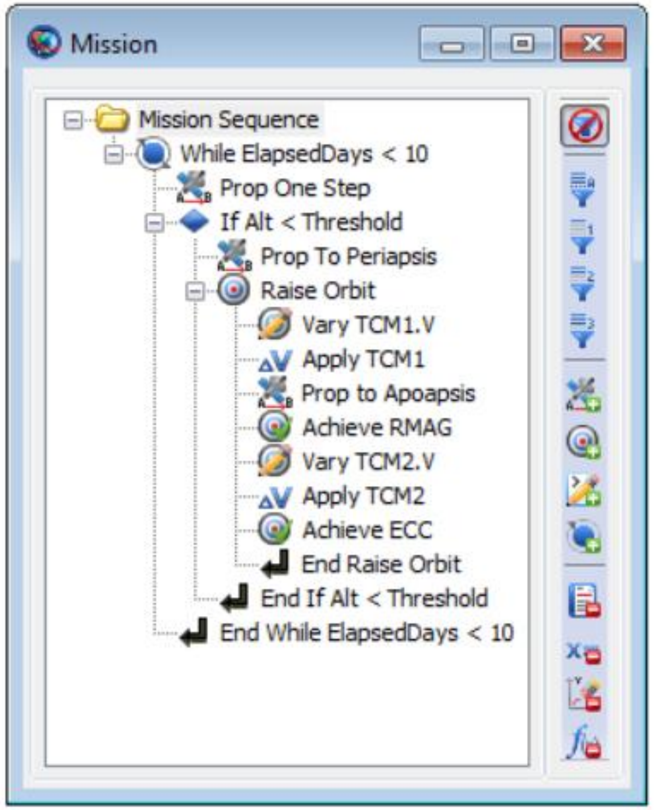


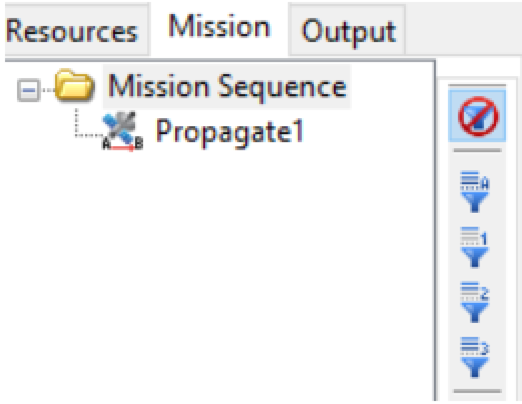
Mission Tab

The Mission tree contain the Mission Sequence, or the exact **sequence of commands** to be executed to model your mission.

Here, on the upper-right image, we can see a complex case in which the mission is composed by lot of commands. Cases like these arise when we try to model some **maneuvers**.

Conversely, in the image at the bottom it is shown the default mission sequence. In general, for many Mission Design tasks, it is sufficient to propagate one orbit. For example, to obtain Eclipse information, ground coverage or detect contact with ground station it is sufficient to have a mission sequence identical to the one in the image in the bottom-right.

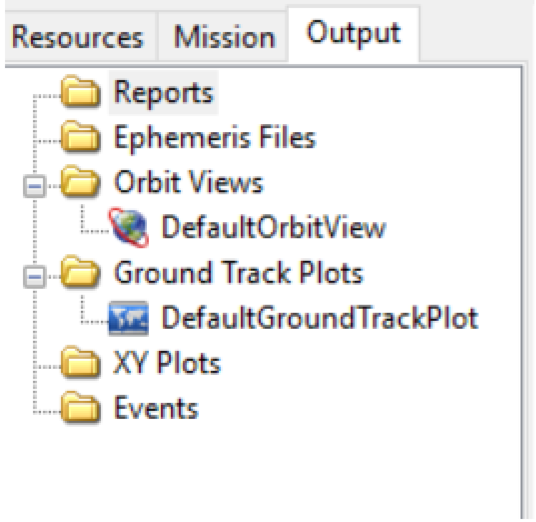




2NDSpace

Output Tab

In the output tab are contained the results of our simulation, or, more precisely, the data file and the plots produced by GMAT using the result of the simulation. Since we can obtain a huge amount of information from the software, it is convenient to be able to select what we want to see. **In other words, we are telling the software what it should display us**.



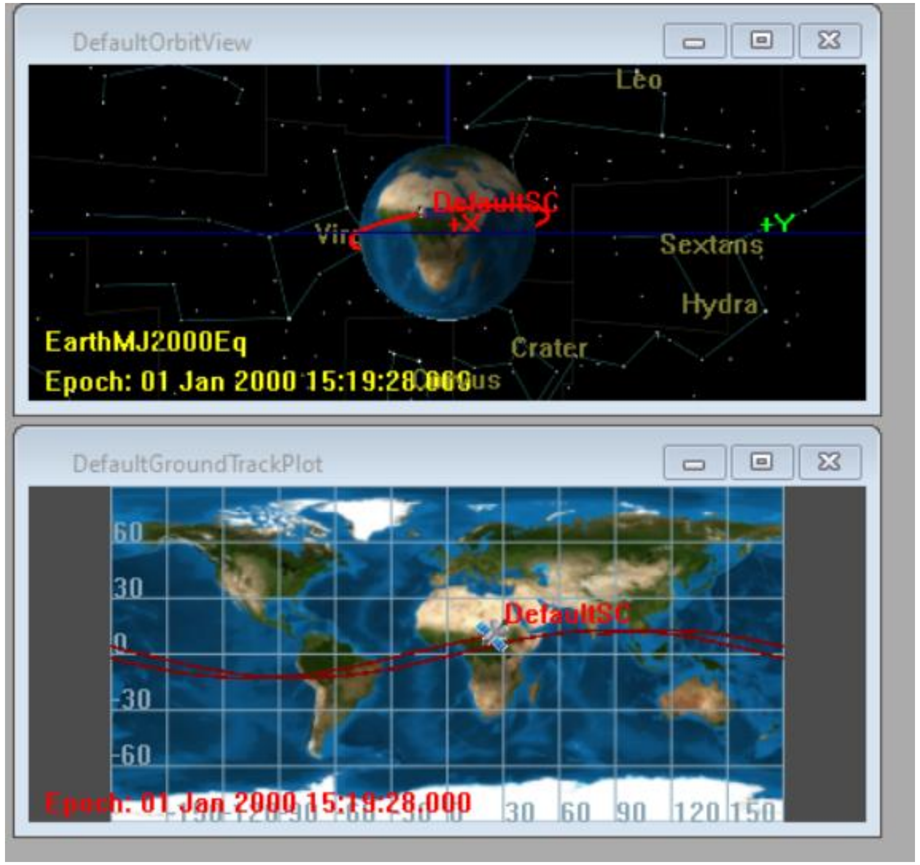
The Output Tab is not editable in the same way as the resource tab: you still have to set-up and edit your outputs in the resource tab, that contains a dedicated output folder. We will see together how to set up new outputs and how to modify the Default ones.

Orbit Viewer and Ground Track Plot

These are two outputs that are included in the default mission.

The first one is an **animated 3D-visualization** of the orbit, while the second is the **Ground Track Plot** of the satellite.

They're really useful: they give the user visual clues about whether the mission has been set up correctly or not.



The Default Mission

What does it mean that we have a Default Mission? Explore the tabs.

In the Resources tab we have:

- A Default Spacecraft, that is obviously necessary for every simulation.
- A Default Propagator, that is the entity that is able to propagate the satellite orbit in time.

In the Mission tab we have:

- A Propagate command, so that the software know to propagate something when you run it.

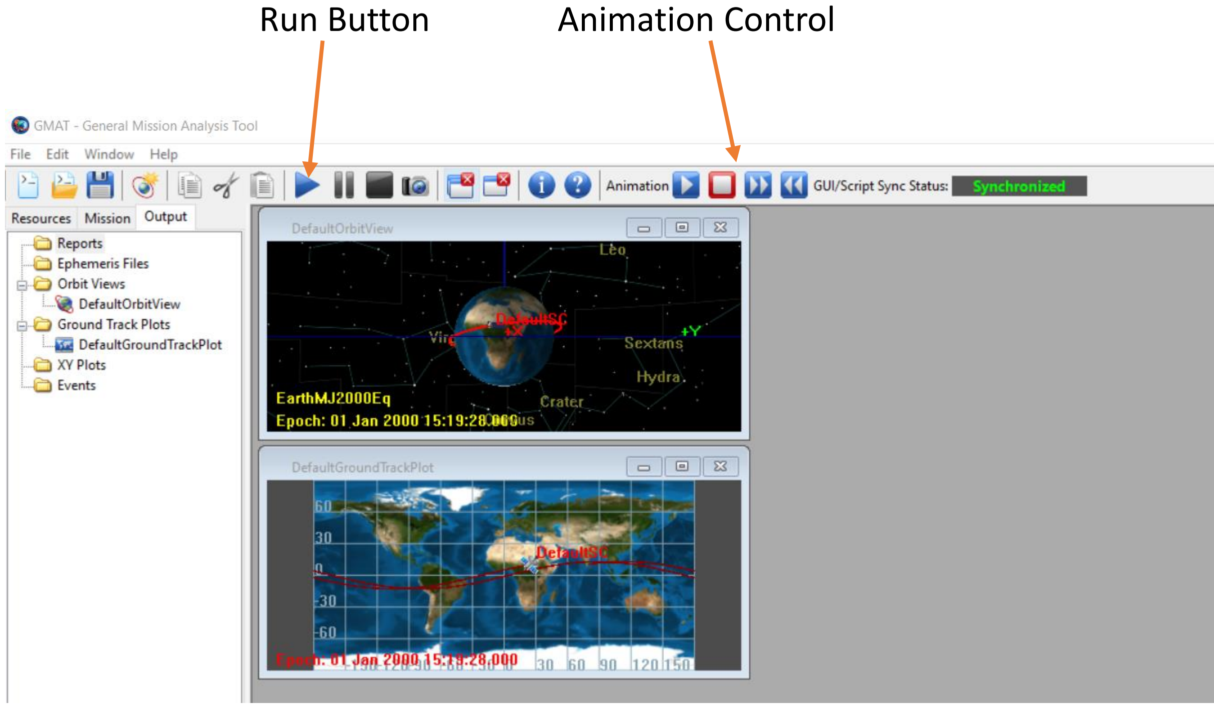
In the Output tab we have:

- A Default Orbit Viewer.
- A Default Ground Track Plot.

Run your first simulation!

Since the Default mission is already loaded, it is easy to run your first space mission: you just need to click the run button!

After the simulation is completed, you can play the animation back by using the animation control commands. You can also slow down the animation if you want to see it better.

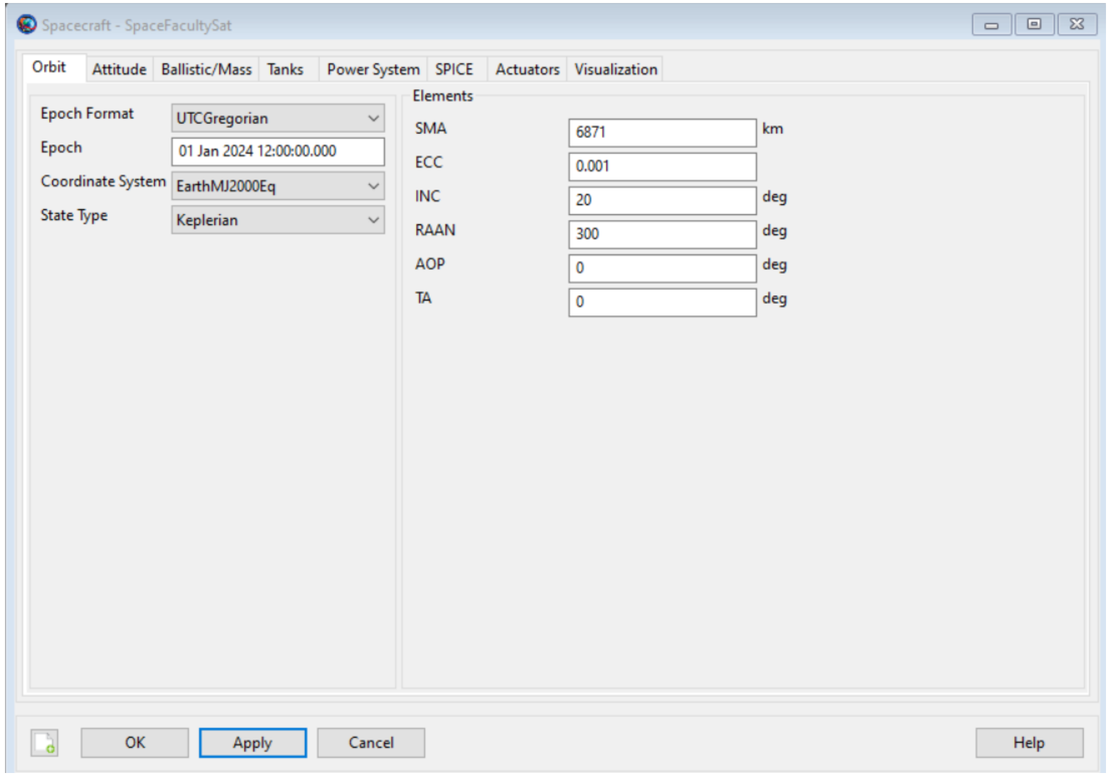


Your first Custom Mission - 1

Now, we can customize the Default Mission.

First of all, right-click on the DefaultSC in the resource tab to rename it. Name your satellite as you want. We will call it **SpaceFacultySat**.

Now double click on the Spacecraft resources and change the parameters as in the image.



Your First Custom Mission - 2

What did we change?

Epoch Format: we have several ways of telling the time and here we have to select one of them. We set the format to UTCGregorian since it is easy to understand and to work with.

Epoch: the time at which our simulation start. We set it to the first of January 2024 at twelve o'clock.

Coordinate system: we also have several ways to describe the position of an object in space. This is not so important for us now, so we keep the default value.

State Type: This is to select the way in which we tell the software the starting position of our satellite. In this case, we will use the Keplerian Elements, since we already talked about them in our lessons.

Elements: these are the values of the Keplerian Elements.

Your first Custom Mission - 3

What are the Keplerian Elements?

- **SMA**: Semi Major Axis. We have seen it during the lesson about orbits: it represents how big our orbit is. In this case, we want to simulate a LEO (Low Earth Orbit) Orbit, so we set this value to 6871 km.

- **ECC**: Eccentricity. We have seen it during the lesson about orbits: it represents how eccentric our orbit is. If this value is close to zero, the orbit is almost circular. We set it to 0.001 to have an almost circular orbit.

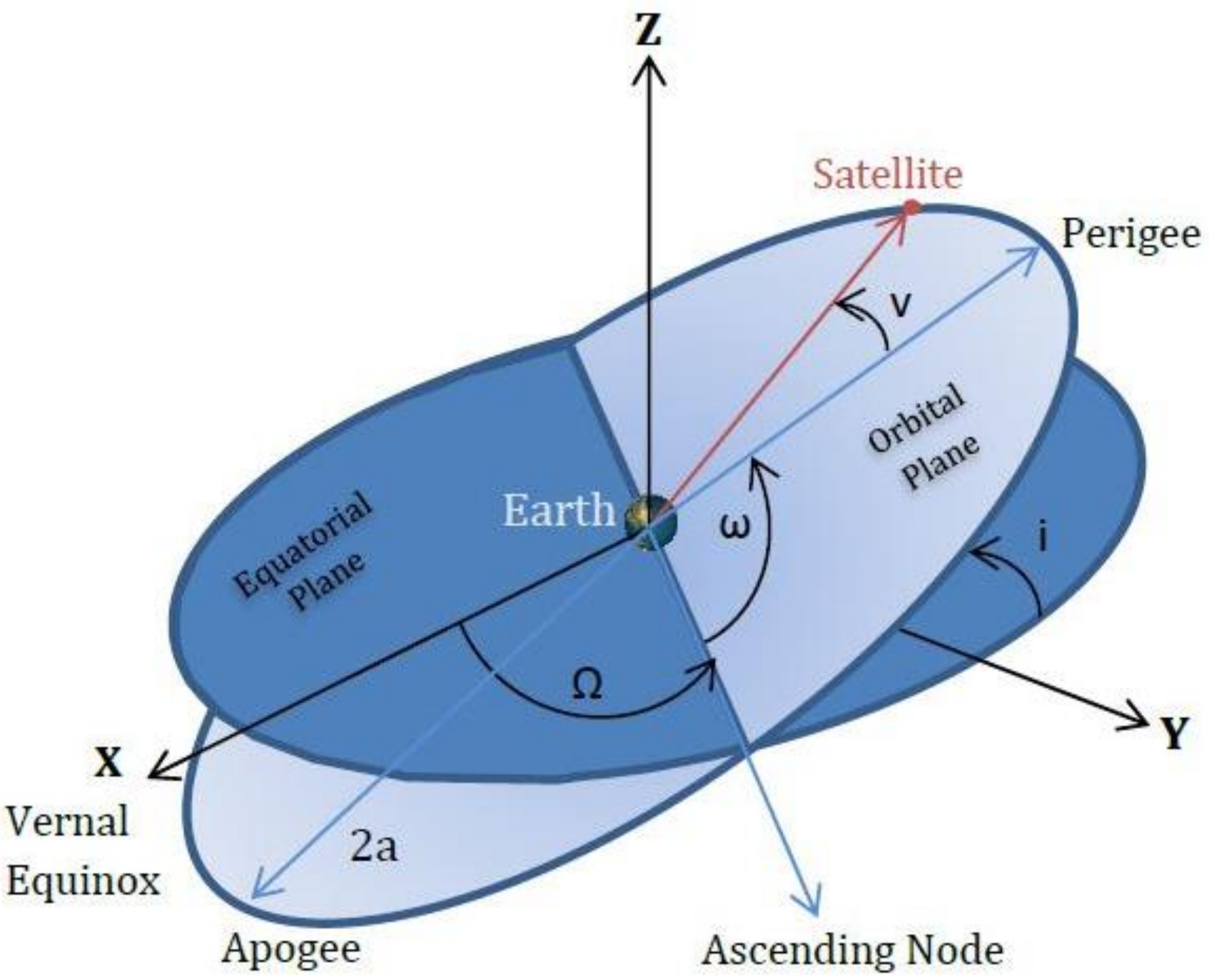
- **INC**: Inclination. We have seen it during the lesson about orbits: it represents how much the orbit is inclined with respect to the equatorial plane. For now, we set this value to 20 deg.

- **RAAN**: Right Ascension of Ascending Nodes. It describes how the orbit is oriented in space. We set it to 300 deg.

- AOP: Argument of Perigee. It describes in which direction the periapsis of the orbit is oriented. We set it to 0, and it is not relevant for the rest of the lesson.

- TA: True Anomaly. It describes at which point of the orbit the satellite is. We set it to 0, and it is not relevant for the rest of the lesson.

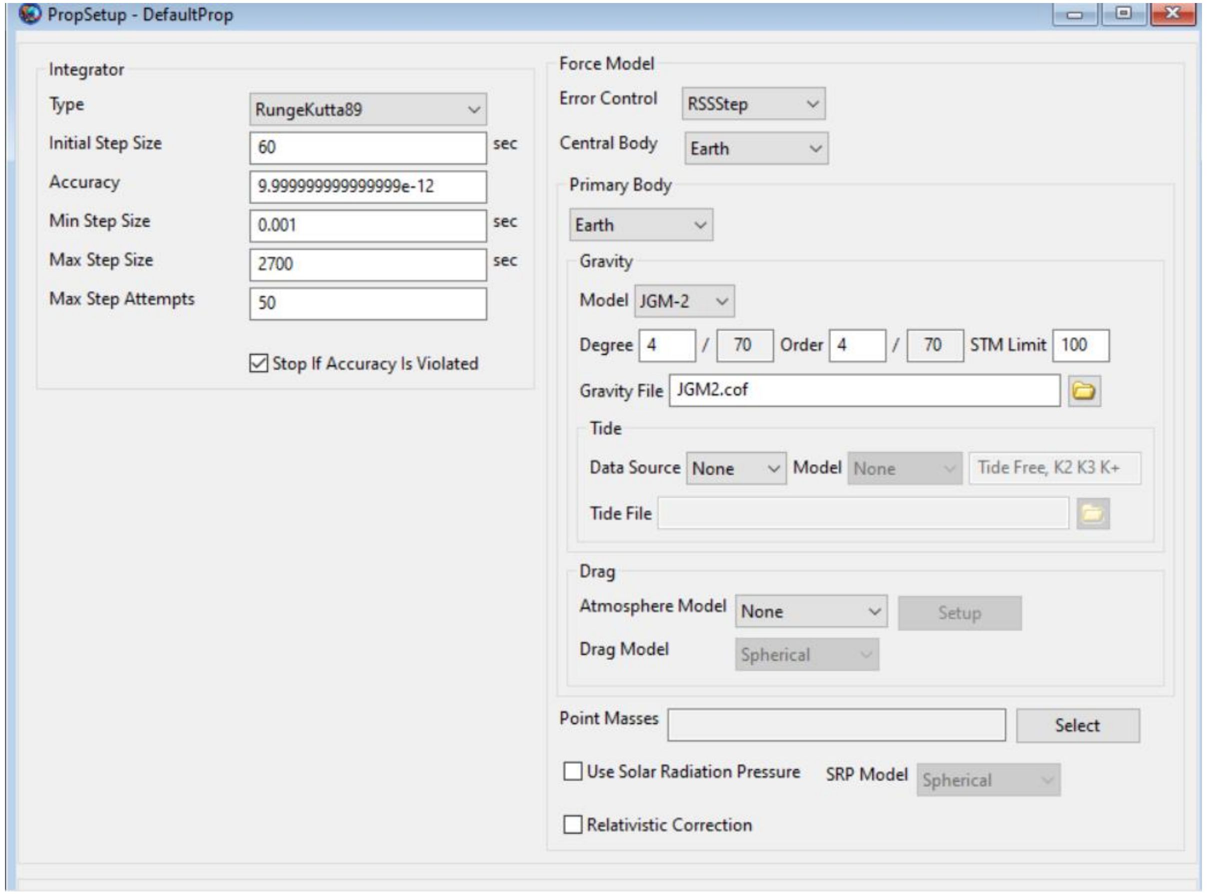
Your first Custom Mission - 3



Your first Custom Mission - 4

Double-clicking the Default Propagator, we can edit also its characteristics.

We won't change anything here, but note that the propagator is able to account for all the perturbations we have seen in our previous lesson. You can see that there is a section for the Drag and one for the Solar Radiation Pressure.



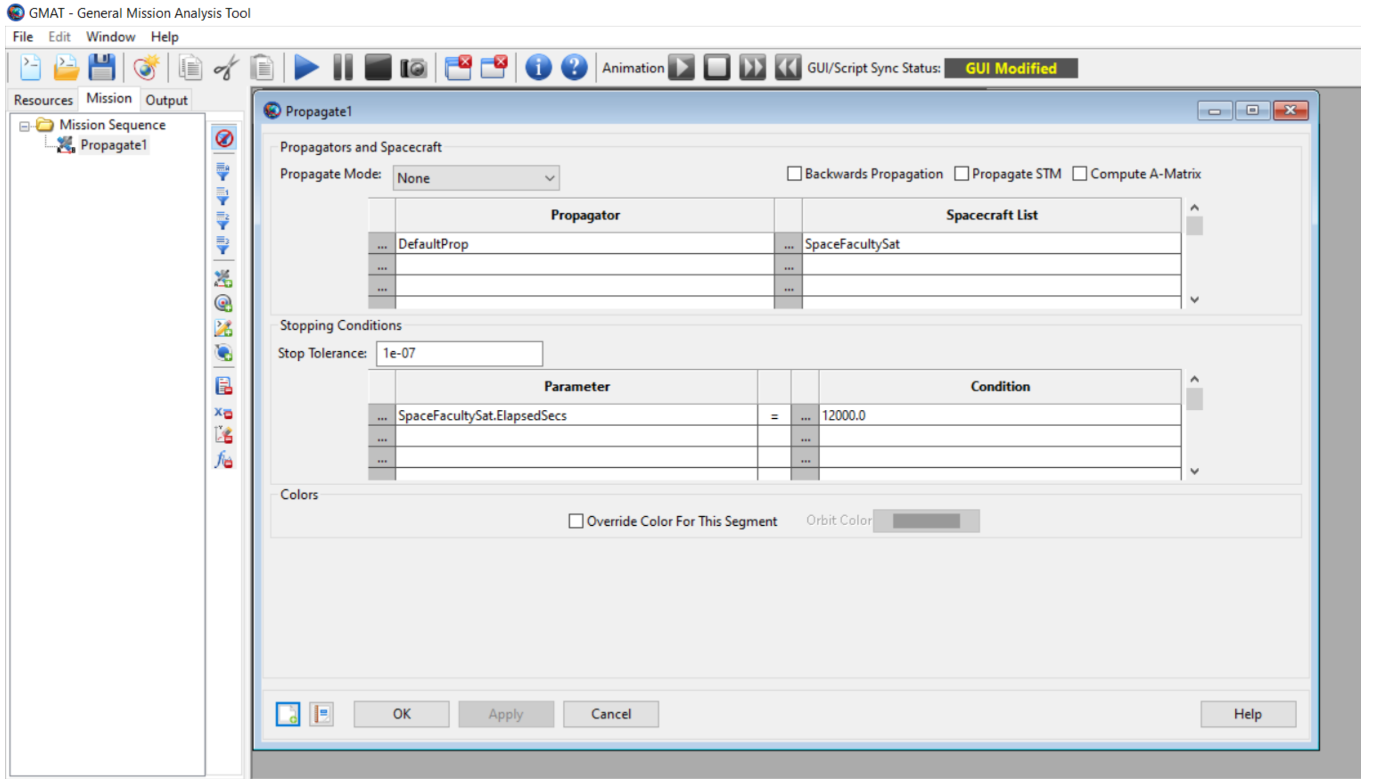
Your first Custom Mission - 5

Now go in the mission tab: we are going to modify also the propagate command. Double click it: you'll see a window that contains two sections.

The first one, *Propagators and Spacecraft*, is the one in which you can assign a spacecraft to one of your existing propagators. We won't do any change here now.

The Second one, *Stopping Conditions*, tells the software when to stop. We want to change it!

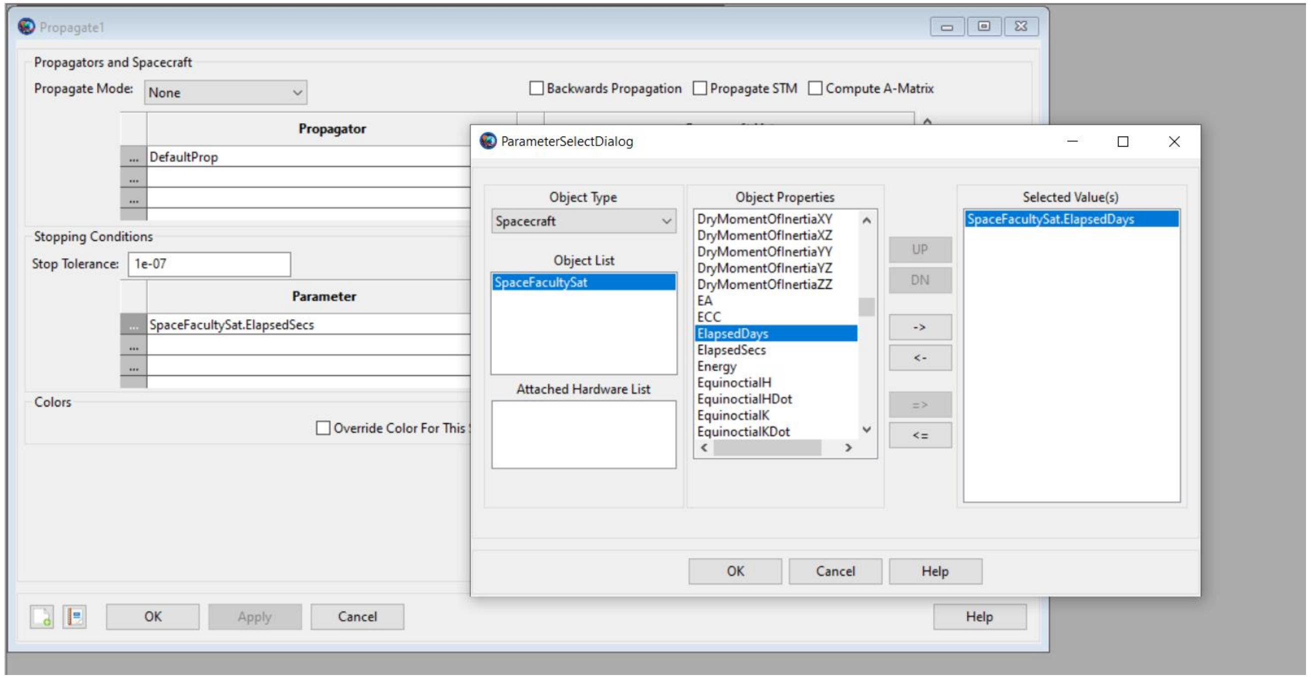
In the column *Parameter* you can decide which parameter will trigger the stop of the simulation. In *Condition* we set at which value of the selected parameter the software should stop. In the Default case, the simulation will stop when the Elapsed Seconds are equal to 12000.



Your first Custom Mission - 6

Click on the three dots on the left of SpaceFacultySat.ElapsedSecs. Doing so, you will open a list of object properties that you can select.

Search for Elapsed Days, double click on it and click ok. Then, type 1 in the Condition column : we want the simulation to stop after one day.

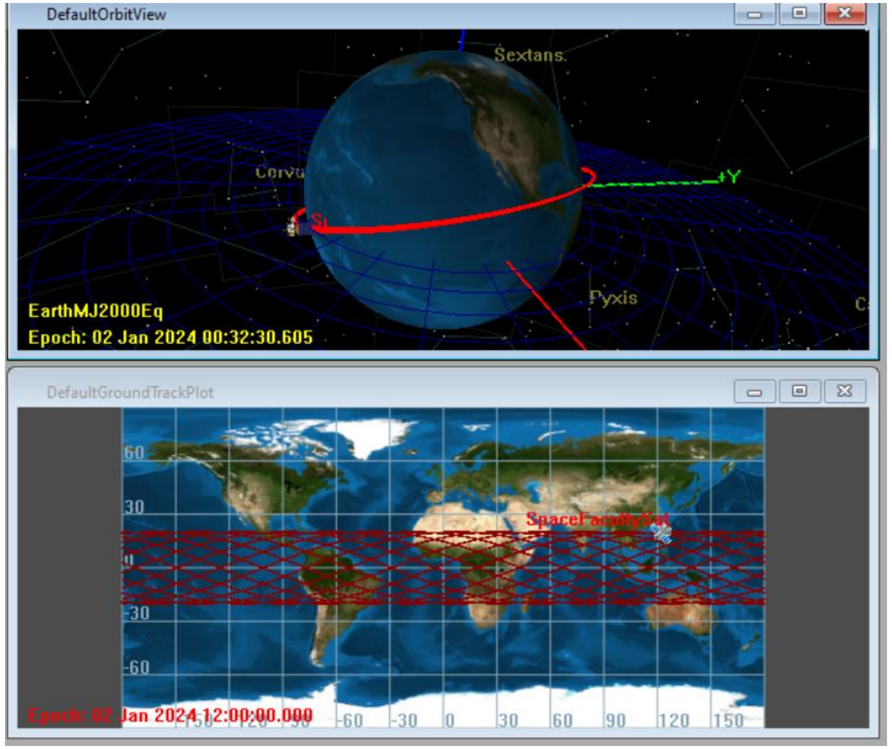


Your first Custom Mission - 7

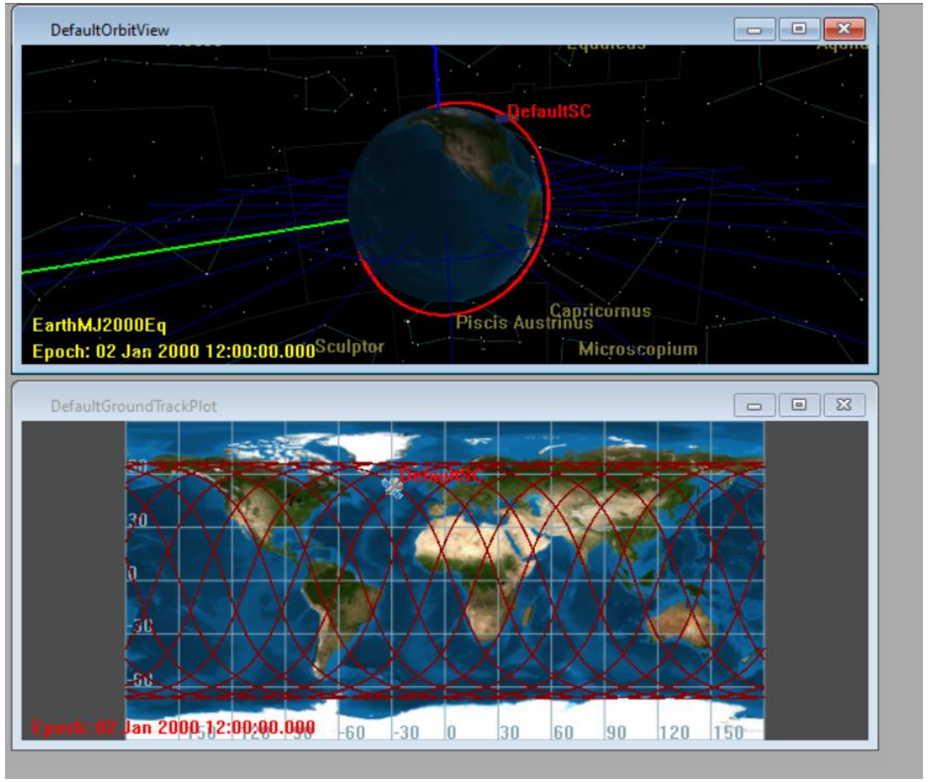
Now Run Your Mission!

You will notice that the orbit is different than the default one, and that you will simulate a lot more orbits, since we told the propagator to go on for an entire day.

If you want to have a better view of your orbit, you can move the 3D Orbit View with your mouse.



Exercise - 1



Since we have talked a lot about Inclination and Eccentricity, try to change the values and see what happens to the orbit. Remember that Inclination values are between 0 and 90 and Eccentricity values are between 0 and they should be less than 1.

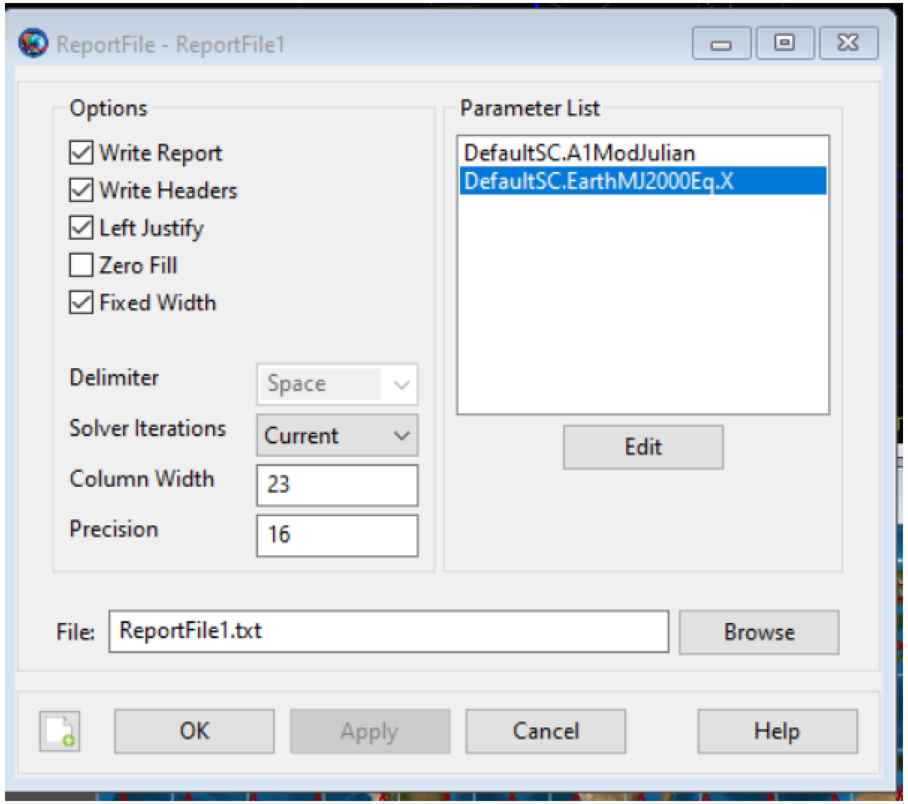
The image on the left is obtained with inclination equal to 60 deg. Try also 0 and 90!

How to get useful results - 1

A lot of the times the results from GMAT are analyzed using other softwares, like MATLAB or Excel.

We can tell the software to give us exactly what we need by adding an output to the simulation.

Go in the resources tab, right click on the output folder and add a report file. A window like the one on the right should open. First, rename the ReportFile as you wish. Then, click edit on the parameter list to decide what we want to save in this file we are creating.

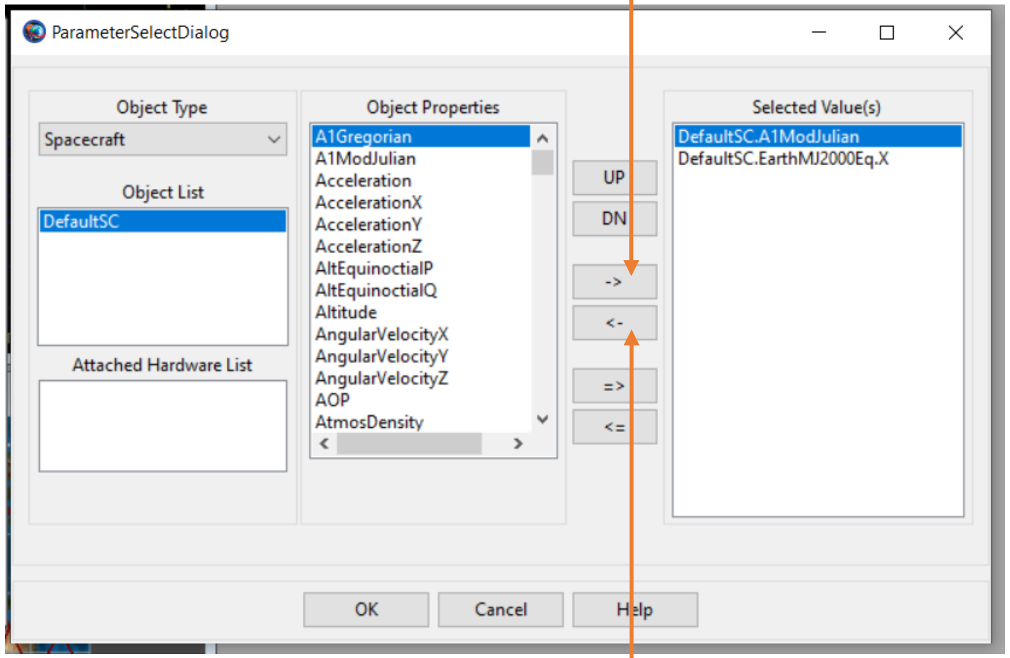


How to get useful results - 2

Now you can select which parameters should be in the Report File. Select the ones already on the right and use the left-pointing arrow button to remove them from the list.

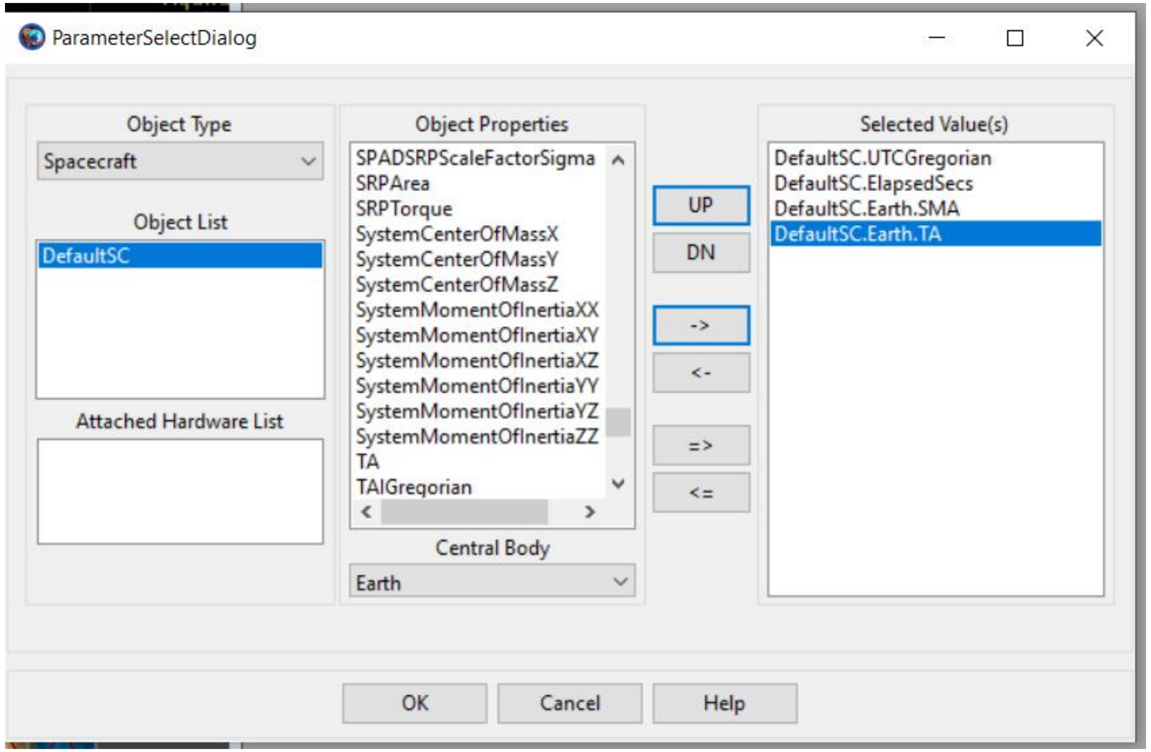
Then, we want to search in the list for UTCGregorian, ElapsedSecs, SMA and TA and add them to the selected values using the right-pointing arrow button.

To add a parameter



To remove a parameter

How to get useful results - 3



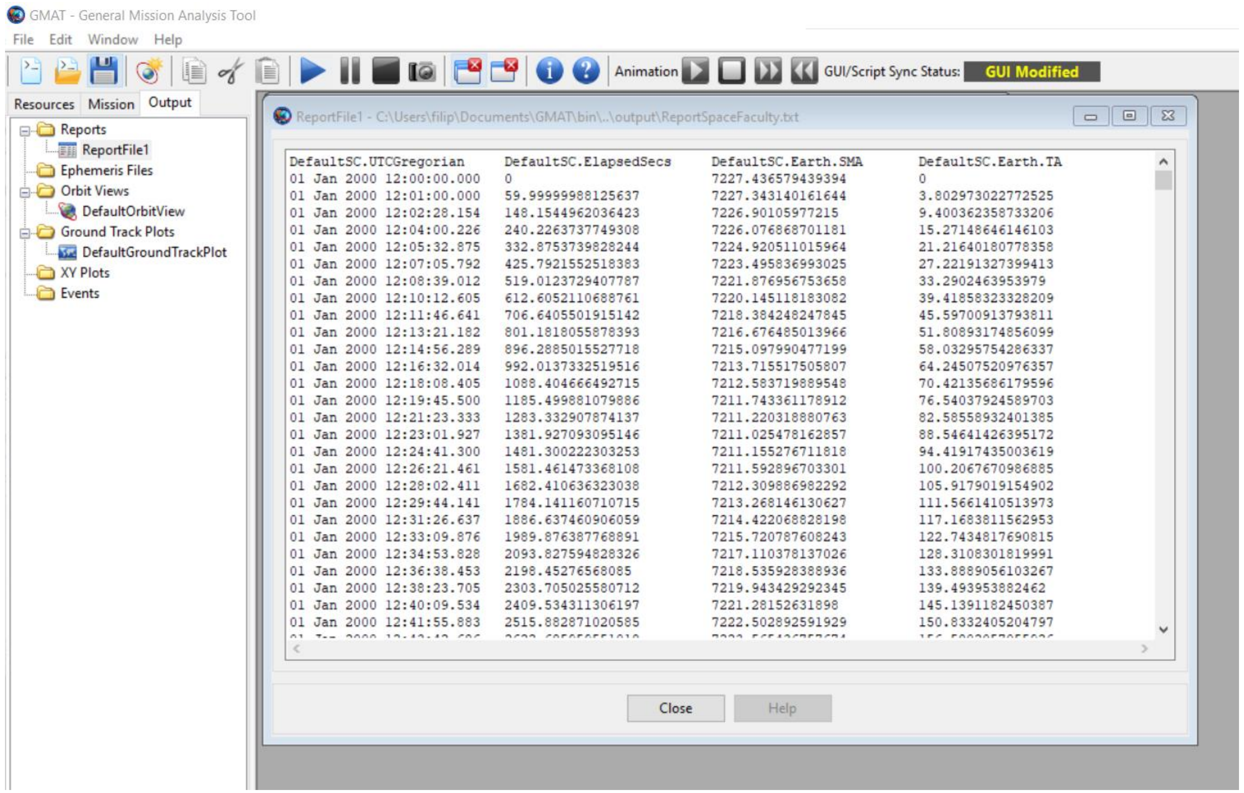
The result should look like in the figure. Click ok, apply the changes and click ok again.

Now, run the simulation again.

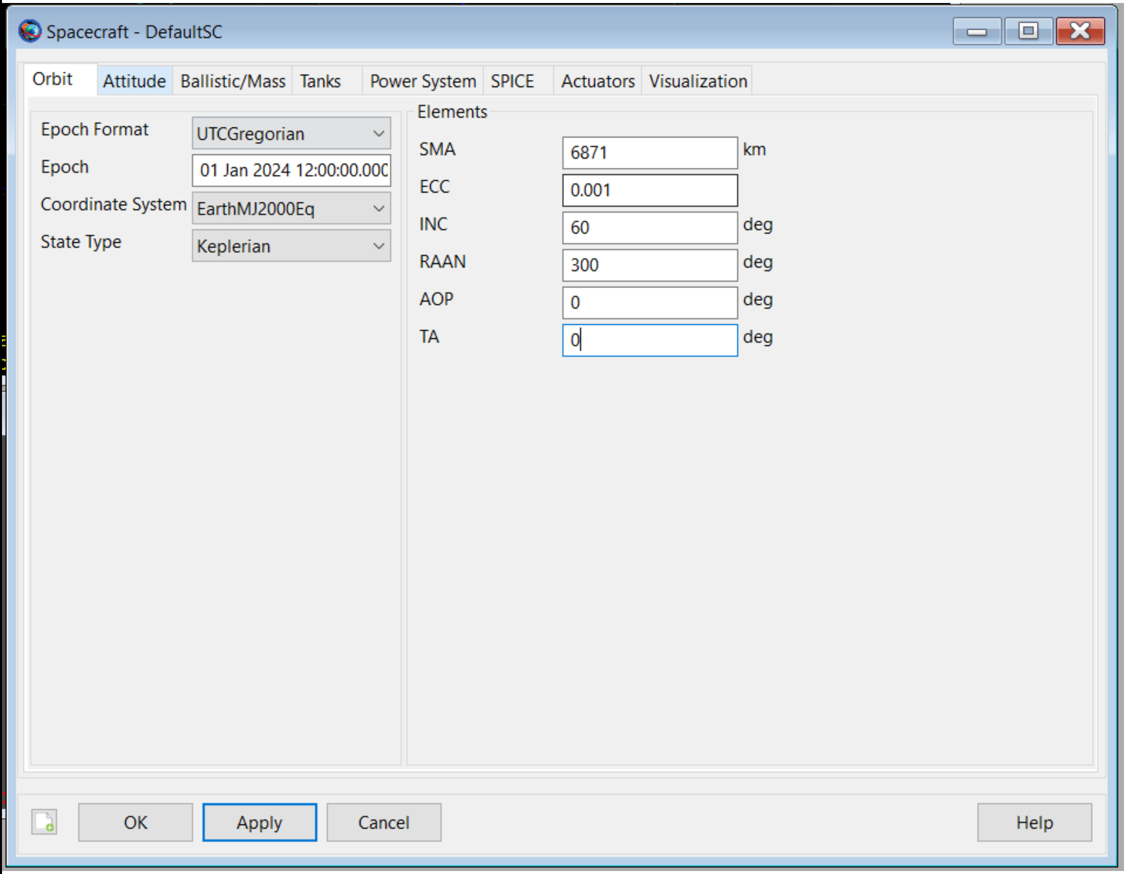
How to get useful results - 4

If you go now in the output tab, you can see that there is something new: a ReportFile!

If you double click on it, you can view the values of the selected parameters. Then, the file is also stored in the GMAT folder, in a sub-folder named output. The file can then be opened and used with lot of third party software.



How to find events - 1



First, we need to be on the same page after all of the experimenting. So, set up the spacecraft to have the same values as in the image on the right.

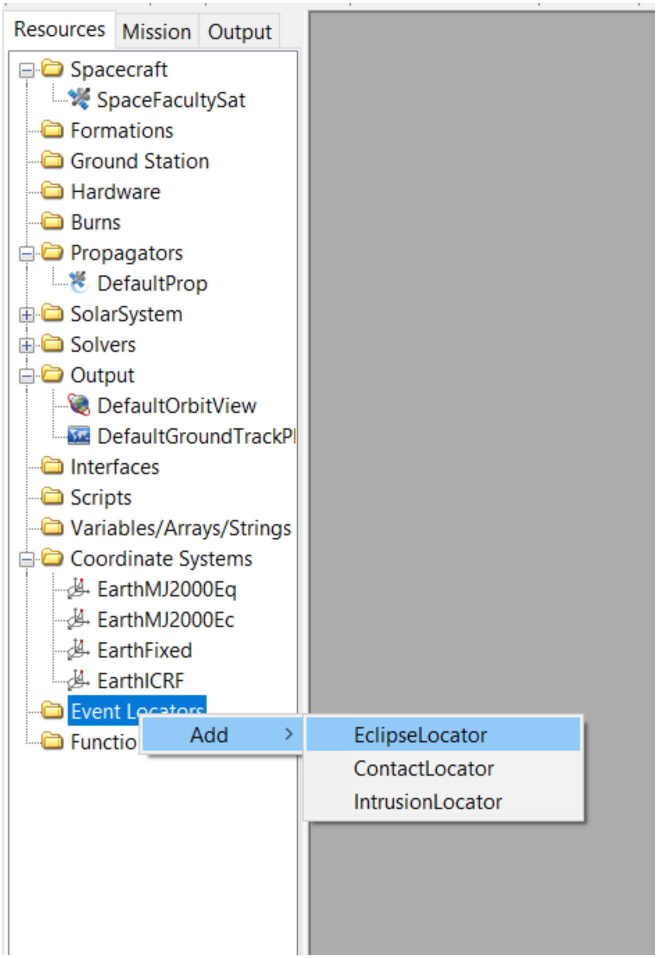
Now, we will see how to locate important events, like eclipse and ground station contacts.

Can anyone remember why eclipses and ground station contacts are important?

How to find events - 2

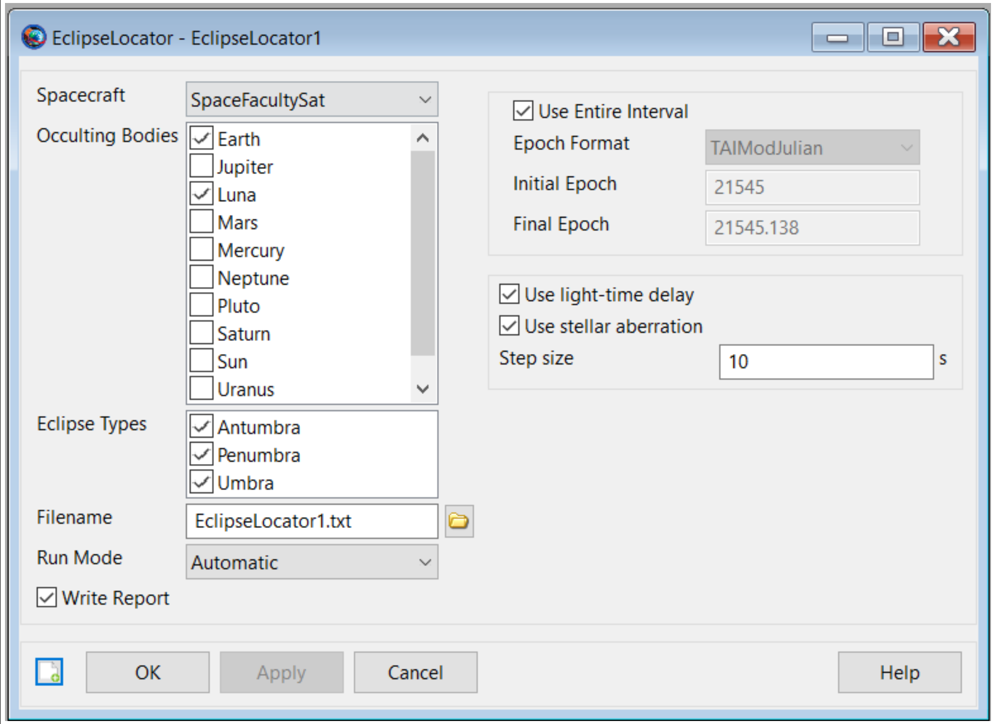
In the Resources Tab, right click on Event Locator, then select an **Eclipse Locator**.

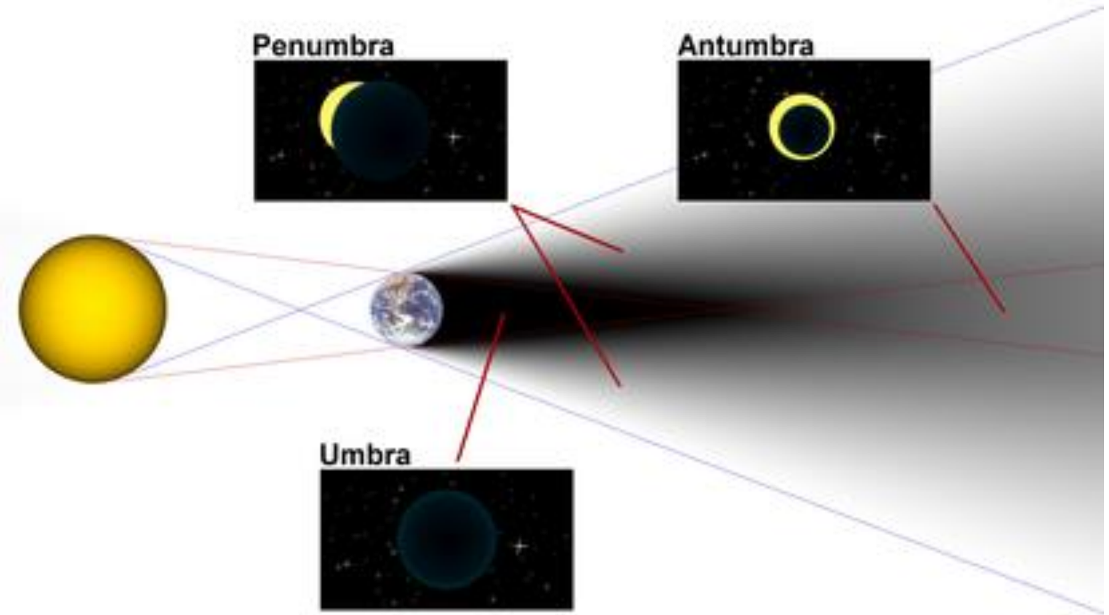
Then repeat the process to add a **contact locator**.



How to find events - 3

We start by setting up the **Eclipse Locator**. Double- Click the Eclipse Locator and check that the values are the same as in the image. As you can see, you can set which occulting bodies to consider. Since we are orbiting around Earth, we need to consider only the shadows cast by Earth and Moon.



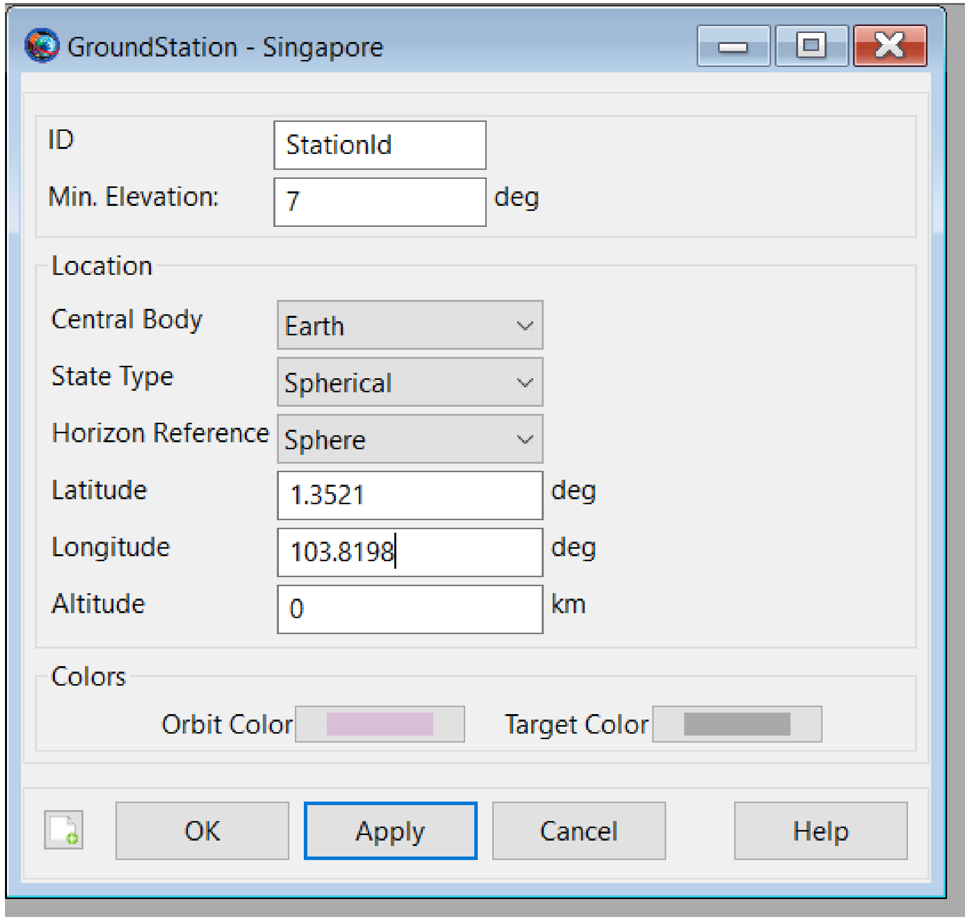


How to find events - 4

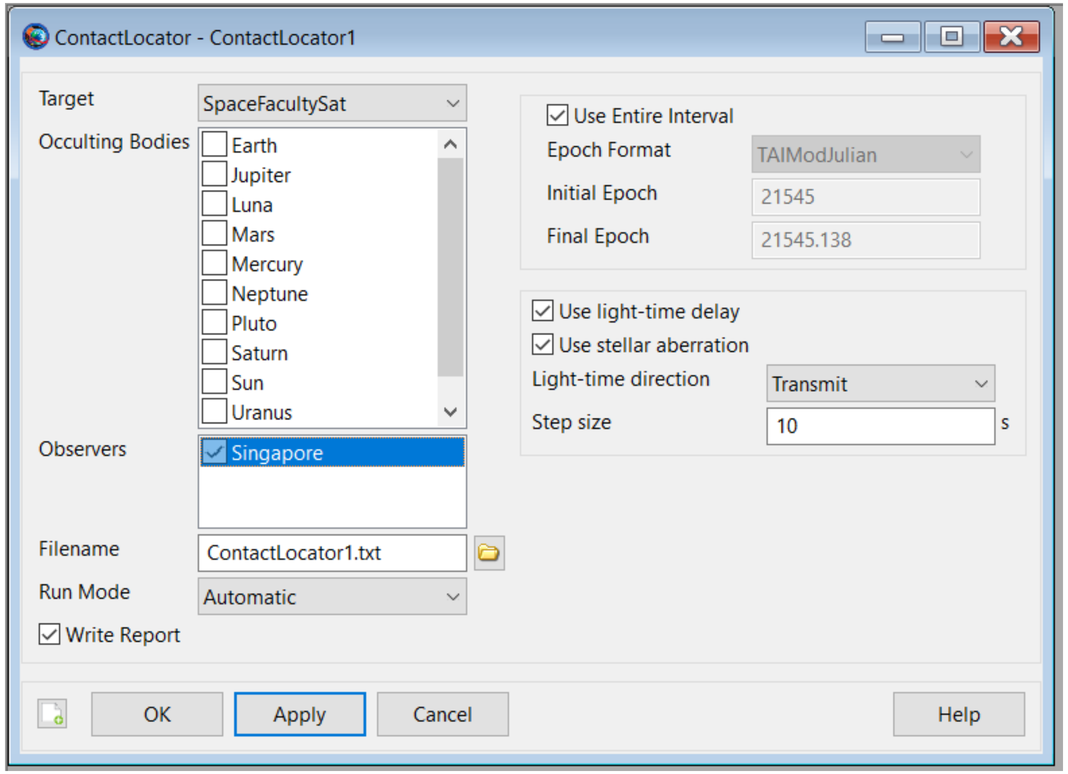
To find a contact event with a ground station, we need first to tell GMAT where our ground station will be.

Right click on the ground station folder in the Resource Tab. Add a new ground station and rename it Singapore.

Now, modify the Horizon Reference to the value Sphere, so we can type in the values of latitude and longitude of Singapore.



How to find events - 5



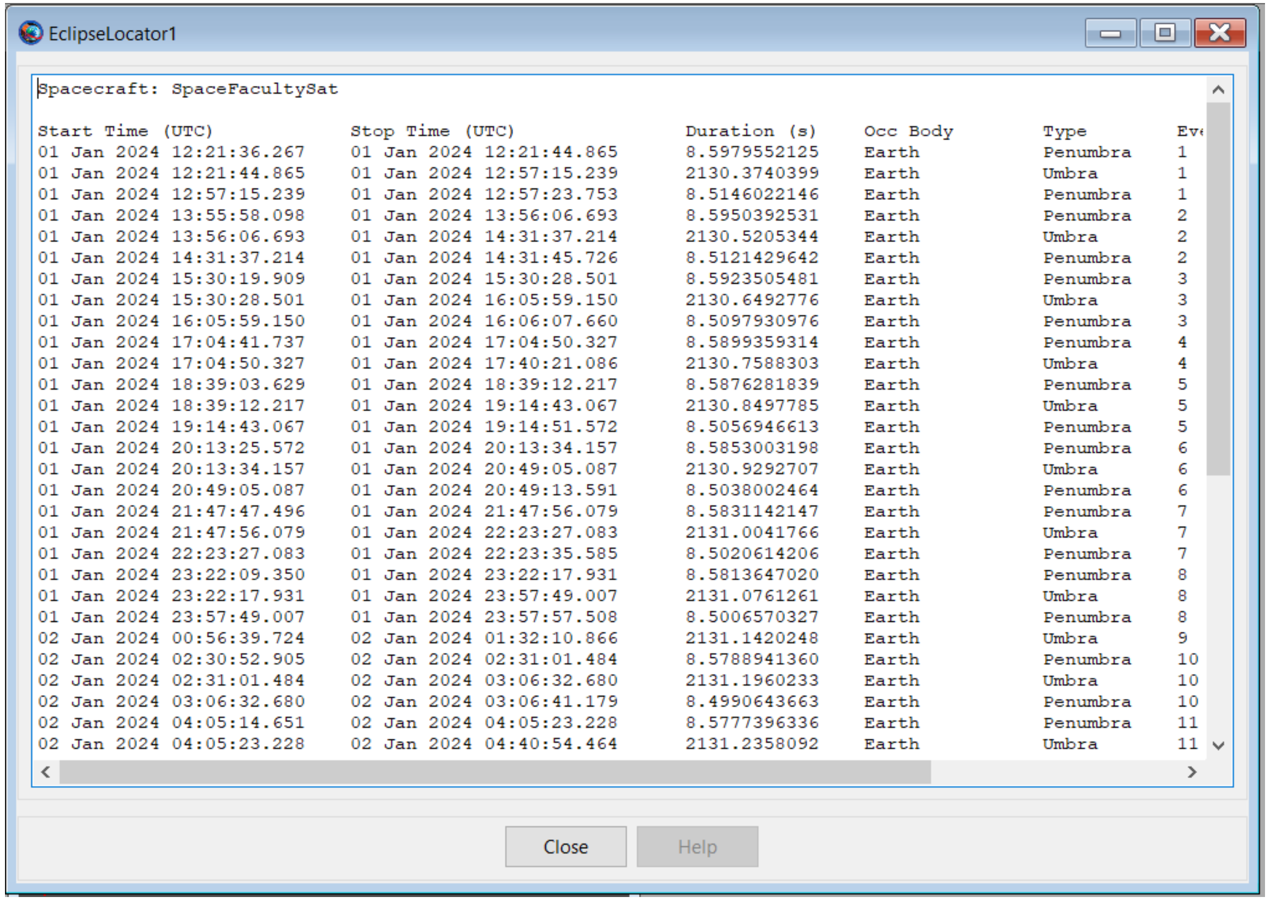
Now we need to set-up the contact locator. We need to select the Singapore Ground Station as an Observer. Note that we do not have to include Earth as an occulting body since the software automatically consider the occultation of the central body of the ground station.

Now you can run the mission again!

How to find events - 6

If you go now to the output tab, you will see that we have two new outputs: an Eclipse Locator and a Contact Locator.

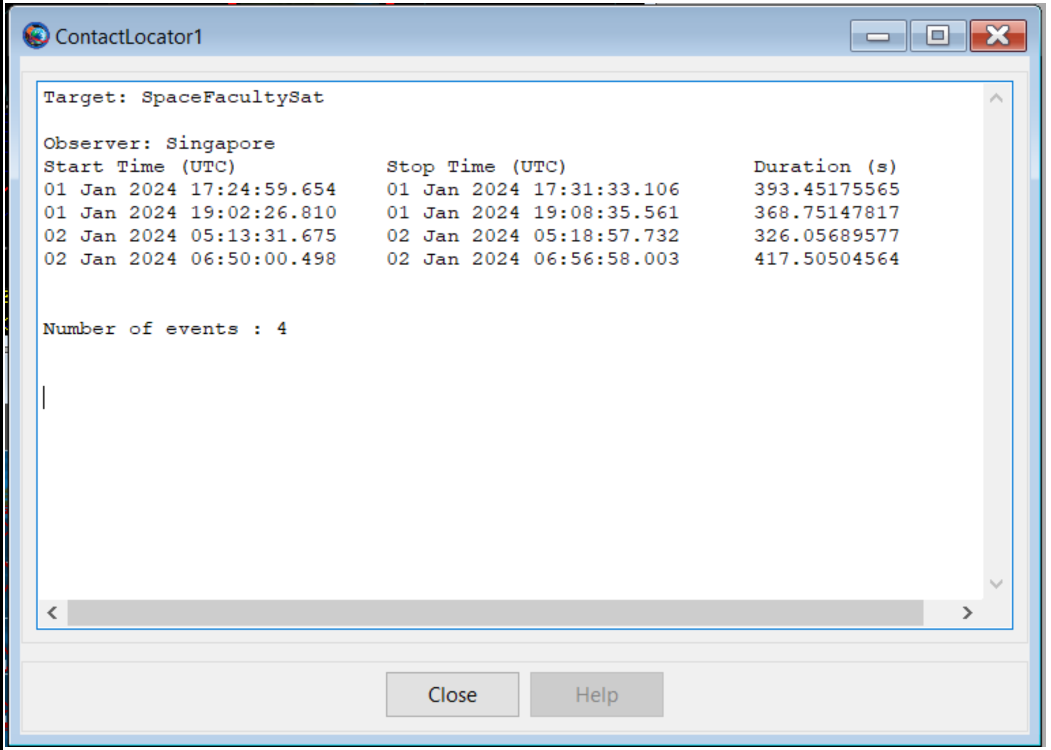
Double Click on the Eclipse Locator. You can see all the eclipse events, when they occurred and their total duration. These consideration are very important for the sake of Power Budget! Can you remember why?



Spacecraft: SpaceFacultySat

Start Time (UTC)	Stop Time (UTC)	Duration (s)	Occ Body	Type	Eve
01 Jan 2024 12:21:36.267	01 Jan 2024 12:21:44.865	8.5979552125	Earth	Penumbra	1
01 Jan 2024 12:21:44.865	01 Jan 2024 12:57:15.239	2130.3740399	Earth	Umbra	1
01 Jan 2024 12:57:15.239	01 Jan 2024 12:57:23.753	8.5146022146	Earth	Penumbra	1
01 Jan 2024 13:55:58.098	01 Jan 2024 13:56:06.693	8.5950392531	Earth	Penumbra	2
01 Jan 2024 13:56:06.693	01 Jan 2024 14:31:37.214	2130.5205344	Earth	Umbra	2
01 Jan 2024 14:31:37.214	01 Jan 2024 14:31:45.726	8.5121429642	Earth	Penumbra	2
01 Jan 2024 15:30:19.909	01 Jan 2024 15:30:28.501	8.5923505481	Earth	Penumbra	3
01 Jan 2024 15:30:28.501	01 Jan 2024 16:05:59.150	2130.6492776	Earth	Umbra	3
01 Jan 2024 16:05:59.150	01 Jan 2024 16:06:07.660	8.5097930976	Earth	Penumbra	3
01 Jan 2024 17:04:41.737	01 Jan 2024 17:04:50.327	8.5899359314	Earth	Penumbra	4
01 Jan 2024 17:04:50.327	01 Jan 2024 17:40:21.086	2130.7588303	Earth	Umbra	4
01 Jan 2024 18:39:03.629	01 Jan 2024 18:39:12.217	8.5876281839	Earth	Penumbra	5
01 Jan 2024 18:39:12.217	01 Jan 2024 19:14:43.067	2130.8497785	Earth	Umbra	5
01 Jan 2024 19:14:43.067	01 Jan 2024 19:14:51.572	8.5056946613	Earth	Penumbra	5
01 Jan 2024 20:13:25.572	01 Jan 2024 20:13:34.157	8.5853003198	Earth	Penumbra	6
01 Jan 2024 20:13:34.157	01 Jan 2024 20:49:05.087	2130.9292707	Earth	Umbra	6
01 Jan 2024 20:49:05.087	01 Jan 2024 20:49:13.591	8.5038002464	Earth	Penumbra	6
01 Jan 2024 21:47:47.496	01 Jan 2024 21:47:56.079	8.5831142147	Earth	Penumbra	7
01 Jan 2024 21:47:56.079	01 Jan 2024 22:23:27.083	2131.0041766	Earth	Umbra	7
01 Jan 2024 22:23:27.083	01 Jan 2024 22:23:35.585	8.5020614206	Earth	Penumbra	7
01 Jan 2024 23:22:09.350	01 Jan 2024 23:22:17.931	8.5813647020	Earth	Penumbra	8
01 Jan 2024 23:22:17.931	01 Jan 2024 23:57:49.007	2131.0761261	Earth	Umbra	8
01 Jan 2024 23:57:49.007	01 Jan 2024 23:57:57.508	8.5006570327	Earth	Penumbra	8
02 Jan 2024 00:56:39.724	02 Jan 2024 01:32:10.866	2131.1420248	Earth	Umbra	9
02 Jan 2024 02:30:52.905	02 Jan 2024 02:31:01.484	8.5788941360	Earth	Penumbra	10
02 Jan 2024 02:31:01.484	02 Jan 2024 03:06:32.680	2131.1960233	Earth	Umbra	10
02 Jan 2024 03:06:32.680	02 Jan 2024 03:06:41.179	8.4990643663	Earth	Penumbra	10
02 Jan 2024 04:05:14.651	02 Jan 2024 04:05:23.228	8.5777396336	Earth	Penumbra	11
02 Jan 2024 04:05:23.228	02 Jan 2024 04:40:54.464	2131.2358092	Earth	Umbra	11

How to find events - 7



If you double-click on the Contact Locator, you can see when the satellite was able to communicate with the ground station and for how long.

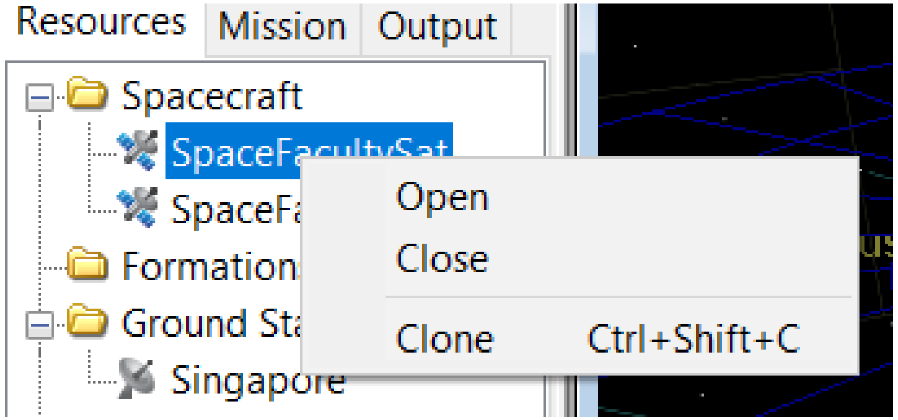
This is important both to send commands to the satellite and to receive data from the spacecraft.

Build your first constellation - 1

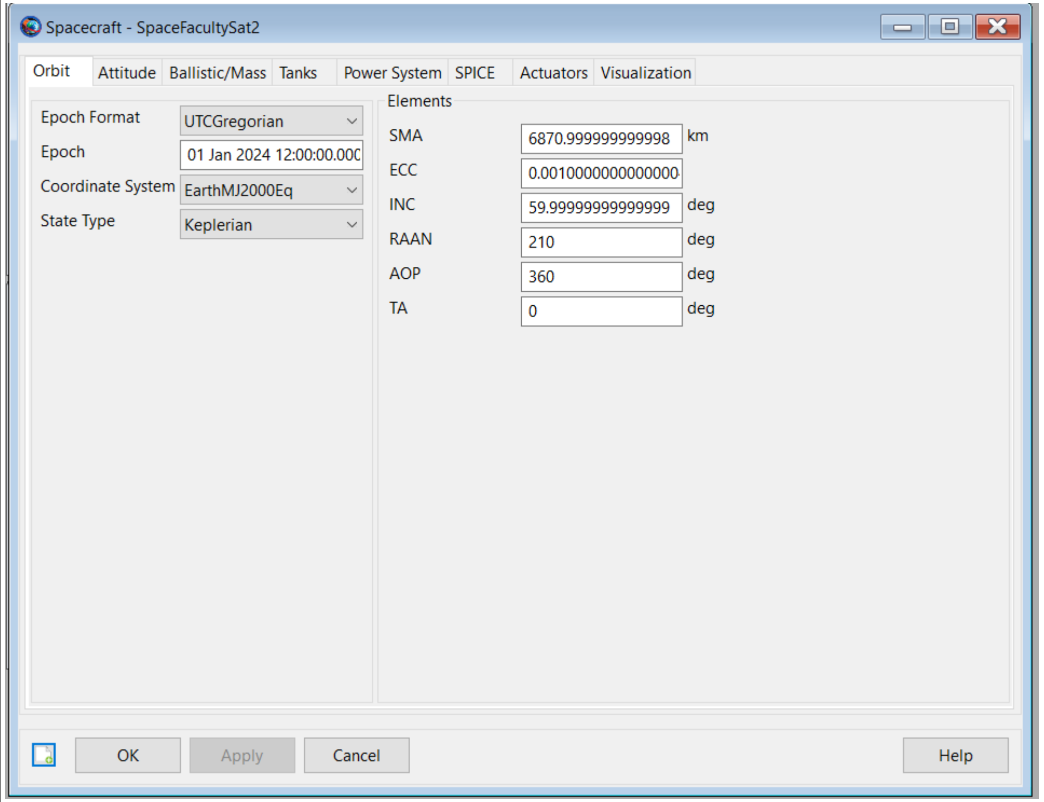
Now that we have made our first mission analysis, it is time to double-up!

We have seen that constellation are very useful in the real world, so we need to learn how to set them up in the software.

First of all, right click on the SpaceFacultySat and click clone. Now you have two spacecrafts!



Build your first constellation - 2



Set the RAAN of the new satellite to be 210 deg.

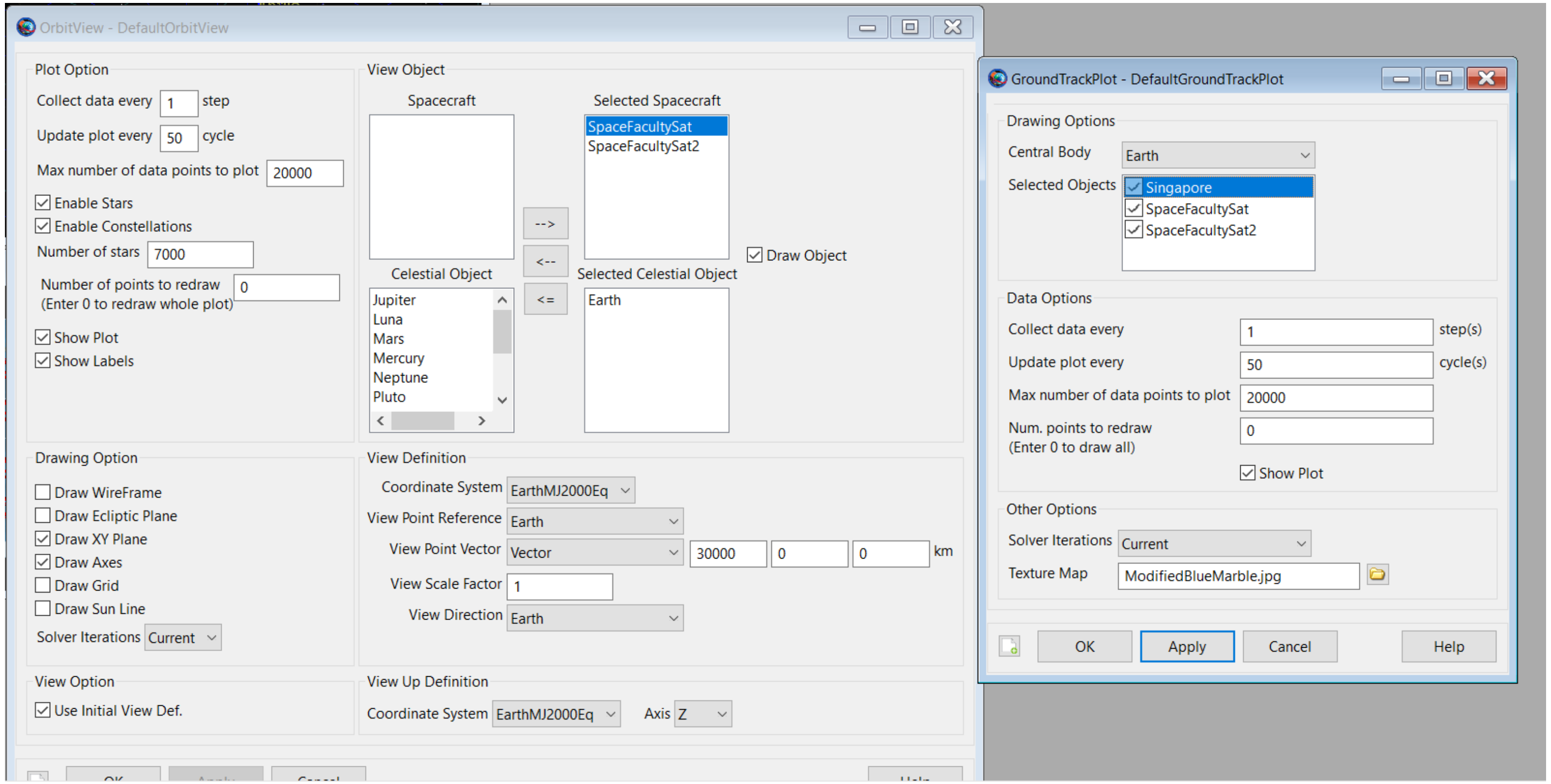
Usually, in real constellation satellites have different RAANs, so that the orbits are in different planes.

If a constellation is made of a great number of satellites, some of them are following each other (at a safe distance!) on the same orbit.

Build your first constellation - 3

If you run the simulation now, you will see no changes even if we have added our satellite. Why? We need to tell GMAT to include it in our output and propagation commands.

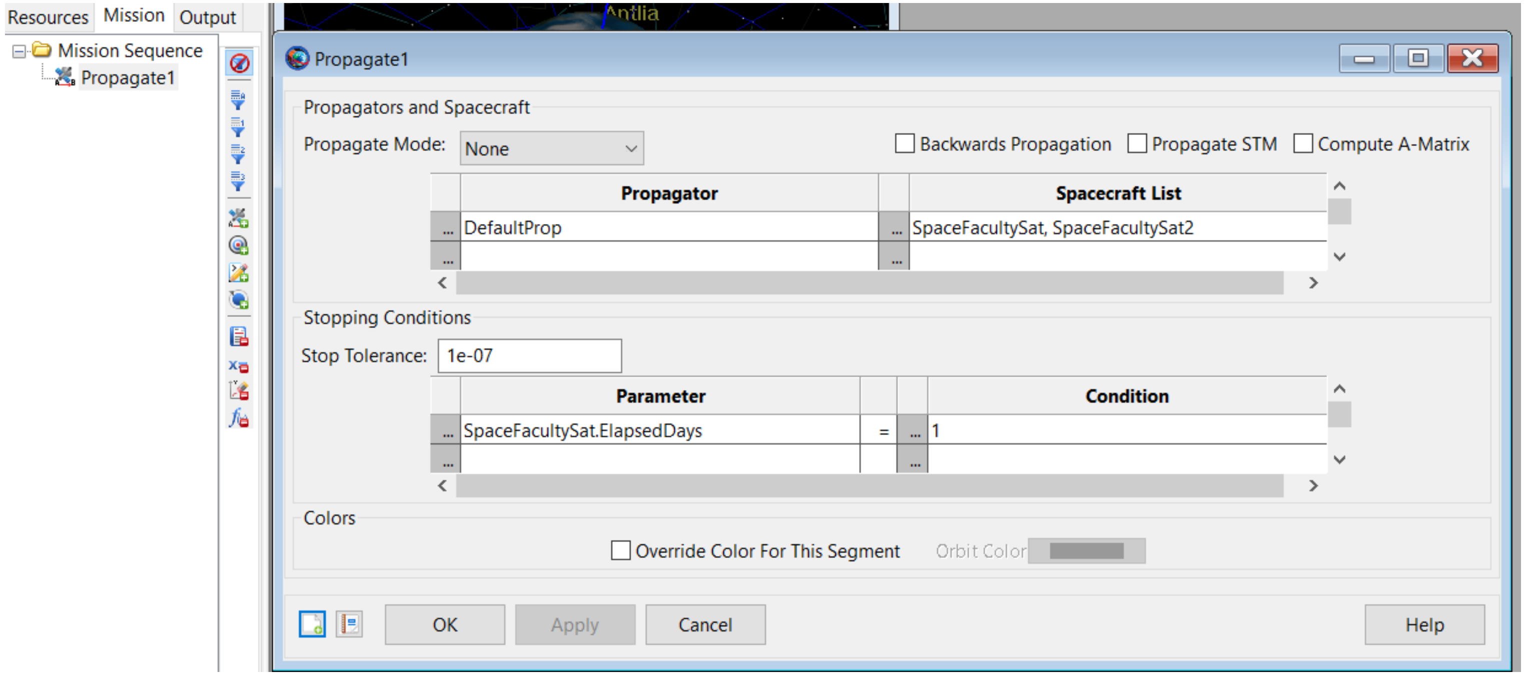
First, edit the DefaultOrbitView and DefaultGroundTrackPlot to include both the spacecrafts.



Build your first constellation - 4

Now we need to tell GMAT that it has also to propagate this new satellite. So, we go in the Mission Tab and we add SpaceFacultySat2 to the Spacecraft List.

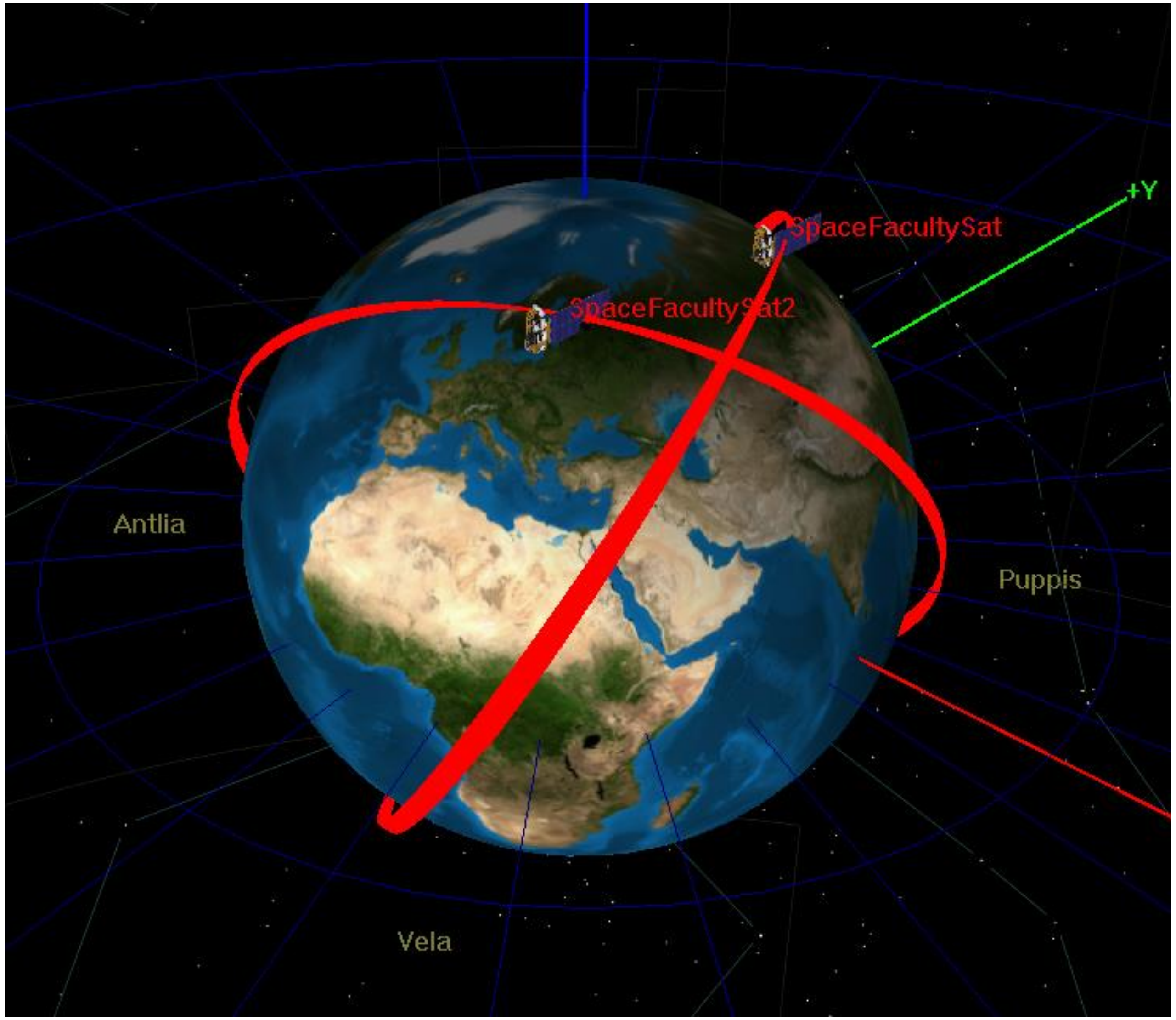
Now everything is ready, and you can Run the mission!



Build your first constellation - 5

Now you simulated also a constellation!

Congrats!



Exercise - 2

- If you try to add the second spacecraft we have created to our Events Locators, you notice that you can't add two of them in a single Locator. Try to set up two more Events Locators for the new satellite, in order to have Eclipse and Ground Contact data also for the new satellite.

- Try to add another Ground Station wherever you want. Note that you can add the new ground station in the event locator!

- Try to add more satellite to the constellation, experimenting with changing RAANs values and TA values.

- Try to set a satellite SMA to 42,164 km and its inclination to 0. What happens to its ground plot? Why?

Thanks for your attention

As we have seen, GMAT is a powerful instrument that you have at your disposal free of charge. It is something that is currently used to do real work out there, so you have the incredible luck to be able to experiment with a software used by professionals.

If you want to learn more, there are a lot of resources out there!

Have fun!

2NDSpace